

Transcendence and undecidability for ℓ -degree Turán densities

Heng Li*

Xizhi Liu†

August 4, 2026

Abstract

We prove that, for every sufficiently large uniformity r and every $1 \leq \ell \leq r - 1$, there is a finite family of r -uniform hypergraphs whose minimum ℓ -degree Turán density is transcendental. For every such fixed pair (r, ℓ) , we also construct a fixed rational threshold $\xi_{r, \ell}$ such that each of the comparisons $\leq$, $=$, and $>$ with $\xi_{r, \ell}$ is undecidable for finite forbidden families, even under the promise that the input density is at least $\xi_{r, \ell}$. The constructions are uniform: partial algorithms guaranteed to halt throughout this range produce the transcendental families and the rational thresholds.

Li, Liu, and Liu prove transcendence and undecidability for ordinary Turán densities. The present paper extends both phenomena to minimum ℓ -degree Turán densities for every $1 \leq \ell \leq r - 1$, including codegree. The main new ingredients are exact finite-family transfers: vertex deletion for $\ell = 1$, a clean-set transfer when $r - \ell \leq \ell$, and marked suspension followed by a two-marker transfer for large $r - \ell$. For transcendence, we use the ordinary construction of Li, Liu, and Liu as input. For undecidability, we instead give an independent rational-power construction, needed because the nonlinear transfers need not send a rational equality value to a rational one. This construction gives fixed rational thresholds after transfer, independently of the input denominator. All resulting forbidden families are explicit finite edge lists, and no reduction evaluates a Turán density.

2020 Mathematics Subject Classification. 05C65, 05C35, 03D35, 68Q17, 11J81.

Keywords. Minimum degree, codegree, Turán density, transcendence, undecidability, rational threshold, hypergraph Lagrangian.

1 Introduction

Let H be an r -uniform hypergraph, or briefly an r -graph. For $S \in \binom{V(H)}{\ell}$, let $d_H(S)$ be the number of edges containing S and let $\delta_\ell(H)$ be the minimum of $d_H(S)$ over all ℓ -sets S . Given a family $\mathcal{F}$ of r -graphs, the corresponding *Dirac–Turán problem* asks for the largest possible $\delta_\ell(H)$ among the $\mathcal{F}$ -free r -graphs of order n . Its normalized asymptotic value is the *minimum ℓ -degree Turán density* $\pi_\ell^{(r)}(\mathcal{F})$.

Throughout the paper, the uniformity in the minimum-degree problem is denoted by r , and we write $s := r - \ell$. When a transfer starts from an ordinary Turán problem of another uniformity, that uniformity is denoted by q ; in statements only about ordinary Turán density, r keeps its usual role. The letter k is used for unrelated finite indices, such as the number of labels in the fixed-order localization argument below.

The case $\ell = 0$ is ordinary Turán density by definition, and the case $\ell = 1$ has the same asymptotic density by a vertex-deletion argument; see [Lemma 2.1](#). At the other endpoint, $\ell = r - 1$ is the codegree density. The parameters $2 \leq \ell \leq r - 2$ interpolate between these endpoints, and their numerical values are often difficult to determine. One may also ask a more structural question: which real numbers occur as densities of *finite* forbidden families, and can exact comparisons between such numbers be decided algorithmically? The surveys of Füredi [\[13\]](#), Sidorenko [\[35\]](#), and Keevash [\[20\]](#) describe the broader hypergraph Turán problem.

For ordinary Turán density, Baber and Talbot [\[2\]](#) and, independently, Pikhurko [\[33\]](#) disproved the old expectation that finite forbidden families might always have rational density. Grosu’s subsequent work [\[15\]](#) developed algebraic and topological operations on sets of Turán densities. Yan and Peng [\[39\]](#) and

*School of Mathematics, Shandong University, Jinan, China. Email: heng.li@sdu.edu.cn.

†School of Mathematical Sciences, University of Science and Technology of China, Hefei, China. Email: liuxizhi@ustc.edu.cn.

Wu [38] later obtained irrational densities for a single forbidden hypergraph, while Liu and Pikhurko [22] constructed finite families whose densities have arbitrarily large algebraic degree.

The distinction between finite and infinite forbidden families is essential here. For possibly infinite families, Pikhurko's construction [33] already implies the existence of transcendental densities, while Liu and Pikhurko [23] showed that the corresponding sets contain intervals. Transcendental constants also occur in related extremal models studied by Mubayi and Terry [31] and Yuster [40], but those constants are not Turán densities of finite forbidden hypergraph families. The finite-family realization problem was recorded in the 2011 AIM problem list of Mubayi, Pikhurko, and Sudakov [30]; it is more restrictive than its infinite-family counterpart.

Minimum-degree densities behave differently. Mubayi and Zhao [32] initiated the systematic study of codegree density. Lo and Markström [24] established the general ℓ -degree setting, proved existence of the limiting density, and showed that for $r > \ell > 1$ the set of ℓ -degree densities is dense in $[0, 1)$. Gao, Pikhurko, Rong, and Sun [14] recently proved that every rational number in $[0, 1)$ is the codegree density of a finite family. These existence results do not address the algorithmic problem studied here: even when a rational target is fixed in advance, deciding equality or either inequality for a given finite forbidden family can be impossible.

Related work of Ai, Ding, Liu, and Yang [1] introduced tree suspensions and explicit transfer functions for single-forbidden ℓ -degree Turán spectra. Their results give infinitely many accumulation points for every $r \geq 3$ and $1 \leq \ell < r$, and densities of arbitrarily large algebraic degree in the ordinary case and in the range $\lceil r/2 \rceil \leq \ell \leq r - 2$. Although both works use suspension constructions, the roles are different: their tree suspensions act within a fixed single-forbidden spectrum, whereas our marked suspension is an auxiliary finite-family construction used before the two-marker transfer when $r - \ell$ is large.

We now state our main results. The first extends finite-family transcendence from ordinary Turán density to minimum ℓ -degree Turán density for every $1 \leq \ell \leq r - 1$, including codegree.

Theorem 1.1. *There is an integer r_0 such that, for every $r \geq r_0$ and every $1 \leq \ell \leq r - 1$, there is a finite family $\mathcal{F}_{r,\ell}$ of r -graphs for which $\pi_\ell^{(r)}(\mathcal{F}_{r,\ell})$ is transcendental. These families can be chosen uniformly: a single partial algorithm halts on every such pair (r, ℓ) and outputs $\mathcal{F}_{r,\ell}$.*

Our second result gives undecidability at a threshold fixed in advance for each pair (r, ℓ) .

Theorem 1.2. *The integer r_0 in Theorem 1.1 may be chosen so that, for every fixed $r \geq r_0$ and $1 \leq \ell \leq r - 1$, there is a rational number $\xi_{r,\ell} \in (0, 1)$ for which each of the following decision problems is undecidable. The input is a finite family $\mathcal{F}$ of r -graphs, and the question is whether*

$$\pi_\ell^{(r)}(\mathcal{F}) \leq \xi_{r,\ell}, \quad \pi_\ell^{(r)}(\mathcal{F}) = \xi_{r,\ell}, \quad \pi_\ell^{(r)}(\mathcal{F}) > \xi_{r,\ell},$$

respectively. Every statement remains true under the promise

$$\pi_\ell^{(r)}(\mathcal{F}) \geq \xi_{r,\ell}.$$

The pair (r, ℓ) and the threshold $\xi_{r,\ell}$ are fixed before the input family is given. Moreover, a single partial algorithm halts on every pair in this range and outputs $\xi_{r,\ell}$.

Here and throughout the paper, a construction is *effective for all sufficiently large parameters* if one partial algorithm halts for every parameter tuple in the stated range. The algorithm need not decide whether an arbitrary tuple lies in that range or take a valid lower bound as input. Our proof gives a computable sufficient bound, but we do not determine the least one.

The fixed rational threshold can be represented by the density of a fixed finite family.

Corollary 1.3. *For every pair (r, ℓ) covered by Theorem 1.2, one can effectively construct a finite family $\mathcal{G}_{r,\ell}$ of r -graphs such that, on input a finite family $\mathcal{F}$ satisfying*

$$\pi_\ell^{(r)}(\mathcal{F}) \geq \pi_\ell^{(r)}(\mathcal{G}_{r,\ell}),$$

it is undecidable whether equality holds or the inequality is strict. Consequently, equality, non-strict order, and strict order comparisons between the minimum ℓ -degree densities of two finite families are undecidable.

The proof uses two ordinary inputs. The fixed-order rational score construction in [Theorem 6.3](#) gives the equality-versus-strict-inequality dichotomy used for undecidability. The rational-power construction in [Theorem 7.1](#) turns its equality value into a rational power that is independent of the input denominator. For transcendence, we use the ordinary theorem of Li, Liu, and Liu [\[21\]](#), stated in [Theorem 6.4](#). This is the only theorem used here without proof.

Proof outline. Put $s := r - \ell$. Once an ordinary input family has been chosen, the passage to minimum ℓ -degree density has three cases.

- (i) If $\ell = 1$, the vertex-degree identity in [Lemma 2.1](#) leaves the ordinary density unchanged.
- (ii) If $\ell \geq 2$ and $1 \leq s \leq \ell$, the clean-set transfer sends a finite family of ℓ -graphs with ordinary density $\alpha > 0$ to a finite family of $(\ell + s)$ -graphs. Its minimum ℓ -degree density is

$$\Theta_s(\alpha) := \frac{\alpha}{(1 + \alpha^{1/s})^s}$$

when $s < \ell$, and is $\alpha/2^\ell$ when $s = \ell$. A clean ℓ -set gives the upper bound, while guard edges give the matching lower construction.

- (iii) If $\ell \geq 2$ and s is large, a marked suspension converts an ordinary family into a one-marker family. The two-marker transfer combines two such families, of ordinary densities a and b , and gives

$$\Omega_s(a, b) := \frac{ab}{(a^{1/s} + b^{1/s})^s}.$$

Two disjoint clean link supports give the upper bound, and a four-part construction gives equality. This transfer does not require a comparison between s and ℓ .

For undecidability, the rational-power construction is used with $h = 1$ in the vertex-degree and large- s cases and with $h = s$ in the bounded- s case. Thus the values obtained from Θ_s and Ω_s are rational. For transcendence, the same transfer maps are strictly increasing and have algebraic inverses, so they preserve transcendence. Every failure of the required clean structure has a bounded witness, and hence every forbidden family can be enumerated effectively without evaluating a Turán density.

The rest of the paper is organized as follows. [Section 2](#) gives the basic definitions and the exact Lagrangian transfer for ordinary Turán density. [Section 3](#) proves the clean-set transfer, [Section 4](#) develops marked suspension, and [Section 5](#) proves the two-marker transfer. [Section 6](#) constructs the ordinary input used for undecidability and states the ordinary transcendence theorem used later. [Section 7](#) gives the rational-power construction. [Section 8](#) proves the main results and the comparison corollary, and [Section 9](#) summarizes the algorithms and their ranges of validity.

2 Basic definitions and the ordinary Lagrangian transfer

2.1 Minimum ℓ -degree Turán density

For an r -graph H and $S \in \binom{V(H)}{\ell}$, the ℓ -link of S is

$$L_H(S) := \left\{ Q \in \binom{V(H) \setminus S}{r - \ell} : S \cup Q \in E(H) \right\}, \quad d_H(S) := e(L_H(S)).$$

The quantity $d_H(S)$ is the ℓ -degree of S . The support of the link is $X_H(S) := \text{supp } L_H(S) := \bigcup_{Q \in E(L_H(S))} Q$, and the minimum ℓ -degree is $\delta_\ell(H) := \min_{S \in \binom{V(H)}{\ell}} d_H(S)$. For a family $\mathcal{F}$ of r -graphs, put

$$\text{ex}_\ell(n, \mathcal{F}) := \max\{\delta_\ell(H) : v(H) = n, H \text{ is } \mathcal{F}\text{-free}\}.$$

When $\ell = 0$, the unique 0-set is empty and $\delta_0(H) = e(H)$. We therefore write $\text{ex}(n, \mathcal{F}) := \text{ex}_0(n, \mathcal{F})$ for the ordinary Turán number. For ordinary Turán density, existence of the limit follows from the averaging argument of Katona, Nemetz, and Simonovits [\[19\]](#). Lo and Markström [\[24\]](#) proved the corresponding statement for every fixed $0 \leq \ell < r$ and every family $\mathcal{F}$: the limit

$$\pi_\ell^{(r)}(\mathcal{F}) := \lim_{n \rightarrow \infty} \frac{\text{ex}_\ell(n, \mathcal{F})}{\binom{n-\ell}{r-\ell}} \quad (2.1)$$

exists. Replacing the denominator by $\binom{n}{r-\ell}$ does not change the value. We write $\pi(\mathcal{F}) = \pi_0^{(r)}(\mathcal{F})$.

The case $\ell = 1$ agrees with ordinary Turán density; see Lo and Markström [24]. We record the identity because it gives the $\ell = 1$ case of the main results.

Lemma 2.1 (Vertex-degree identity). *For every $r \geq 2$ and every family $\mathcal{F}$ of r -graphs,*

$$\pi_1^{(r)}(\mathcal{F}) = \pi(\mathcal{F}).$$

Deleting isolated vertices is a standard reduction in Turán problems; see, for example, Keevash [20]. We record the finite-family form used below.

Lemma 2.2. *Let $\mathcal{A}$ be a finite family of r -graphs. For $A \in \mathcal{A}$, let A° be obtained by deleting all isolated vertices, and remove duplicates from $\mathcal{A}^\circ := \{A^\circ : A \in \mathcal{A}\}$. Then $\pi(\mathcal{A}^\circ) = \pi(\mathcal{A})$. Moreover, for every sufficiently large n , an n -vertex r -graph is $\mathcal{A}$ -free if and only if it is $\mathcal{A}^\circ$ -free.*

2.2 Hypergraph Lagrangians and exact ordinary transfer

The weighting method goes back to Motzkin and Straus [29] and was developed for hypergraphs by Frankl and Füredi [10] and Talbot [37]. The *Lagrangian* of an r -graph G is

$$p_G(x) := \sum_{e \in E(G)} \prod_{v \in e} x_v, \quad \lambda(G) := \max \left\{ p_G(x) : x_v \geq 0, \sum_v x_v = 1 \right\}.$$

The *class Lagrangian* of a class $\mathcal{D}$ of r -graphs is $\Lambda(\mathcal{D}) := \sup_{G \in \mathcal{D}} \lambda(G)$. Blowups of G with class proportions tending to x have edge density tending to $r!p_G(x)$; Brown and Simonovits [8] developed the role of blowups in extremal hypergraph densities. Pikhurko [33] showed, in particular, that $r!\lambda(G)$ can be realized as the Turán density of a finite family for every finite r -graph G .

An r -graph *covers pairs* if every pair of vertices lies in an edge. If A is an r -graph, an *A -pair-covering extension* is an r -graph Q containing an injective copy of A such that every pair of vertices in the distinguished copy lies in an edge of Q . It is enough to retain extensions on at most

$$M(A) := v(A) + (r-2) \binom{v(A)}{2}$$

vertices. Let $\text{Ext}(A)$ contain one representative of every such extension on at most $M(A)$ vertices.

The following minimum-support fact for hypergraph Lagrangians is standard; see Frankl and Rödl [11].

Lemma 2.3. *Let G be an r -graph with $\lambda(G) > 0$. Choose a maximizing weighting of minimum support and let D be the induced subgraph on its positive support. Then $\lambda(D) = \lambda(G)$ and D covers pairs.*

The following is the standard extension argument relating Lagrangian density to ordinary Turán density; see Sidorenko [36] and Keevash [20, Section 3]. We record the finite-family form used below. Its finiteness and effective enumerability follow directly from the definition of $\text{Ext}(A)$.

Theorem 2.4. *Let $\mathcal{D} = \text{Forb}(\mathcal{A})$ be a monotone class of r -graphs with $\mathcal{A}$ finite, and put*

$$\mathcal{F}(\mathcal{A}) := \bigcup_{A \in \mathcal{A}} \text{Ext}(A).$$

Then $\mathcal{F}(\mathcal{A})$ is finite and effectively enumerable, and

$$\pi(\mathcal{F}(\mathcal{A})) = r!\Lambda(\mathcal{D}).$$

3 The clean-set transfer when $s \leq \ell$

Fix integers $\ell \geq 2$ and $1 \leq s \leq \ell$, and put $r = \ell + s$. Let $\mathcal{A}$ be a finite family of ℓ -graphs with $\alpha := \pi(\mathcal{A}) > 0$. By Lemma 2.2, we assume that every member of $\mathcal{A}$ has no isolated vertex.

The hypotheses have separate roles. Finiteness of $\mathcal{A}$ makes the certificate family finite, while the removal of isolated vertices ensures that every vertex of a forbidden link copy is witnessed by a prescribed edge. The numerical value α is not an input to the certificate construction; it enters only the asymptotic analysis. Condition (C1) below controls edges meeting a link support too heavily, whereas (C2) imports the ordinary Turán bound into restricted links. The cases $s < \ell$ and $s = \ell$ share the same certificates and upper-bound argument, but require different lower constructions.

3.1 Clean sets and the upper bound

We begin with the local condition that will certify the upper bound.

Definition 3.1. Let H be an r -graph and $S \in \binom{V(H)}{\ell}$. Put $X := X_H(S) = \text{supp } L_H(S)$. We call S $\mathcal{A}$ -clean if the following two conditions hold:

- (C1) every edge $E \in E(H)$ satisfies $|E \cap X| \leq \ell$;
- (C2) for every $Q \in \binom{V(H)}{s}$, the ℓ -graph $L_H(Q)[X]$ is $\mathcal{A}$ -free.

In the upper-bound calculation, (C1) forces every completion of an ℓ -set inside X to leave X , and (C2) bounds each resulting restricted link by $\text{ex}(|X|, \mathcal{A})$.

We now turn this definition into a finite ordinary forbidden family. We use the following quotient convention throughout the paper.

Definition 3.2. A finite positive configuration consists of finitely many formal vertices, a finite list of prescribed r -sets, and finitely many tuples required to be injective. An admissible quotient is obtained from an equivalence relation on the formal vertices such that every prescribed r -set still has r distinct equivalence classes and every tuple declared injective remains injective. Its edges are the images of the prescribed r -sets, with repeated images merged.

There are only finitely many admissible quotients of a fixed finite configuration, and they can be enumerated effectively. Whenever a formal copy is called *injective*, its full vertex tuple is among the tuples required to remain injective; the same convention applies to all explicitly marked distinct vertices. Except for incidences and identifications explicitly prescribed, every formal symbol introduced below is initially a fresh vertex. All additional coincidences that may occur in a host embedding are represented by the admissible quotients.

For a formal ℓ -set S , define two kinds of certificates.

Type I certificate. Take a formal r -edge E , choose distinct marked vertices $z_1, \dots, z_{\ell+1} \in E$, and, for each i , take an s -set P_i containing z_i and disjoint from S . Prescribe the edges

$$E, \quad S \cup P_i \quad (1 \leq i \leq \ell + 1).$$

This certifies an edge containing at least $\ell + 1$ vertices of $X_H(S)$.

Type II certificate. Choose $A \in \mathcal{A}$, a formal s -set Q , and an injective copy $U = \{u_a : a \in V(A)\}$ disjoint from Q . Prescribe

$$Q \cup \{u_a : a \in e\} \quad (e \in E(A)).$$

For every $a \in V(A)$, also choose an s -set P_a containing u_a and disjoint from S , and prescribe

$$S \cup P_a.$$

This certifies a copy of A in some $L_H(Q)[X_H(S)]$; the latter edges explicitly certify membership of every copy vertex in the support $X_H(S)$.

Take ℓ pairwise disjoint formal r -edges $R_1, \dots, R_\ell$, and put $R := R_1 \cup \dots \cup R_\ell$. Their union is only a finite reservoir of candidate ℓ -sets; the lower bound on $\delta_\ell(H)$ will guarantee that an actual host contains such disjoint root edges. For every $S \in \binom{R}{\ell}$, choose independently one Type I or Type II certificate rooted at S . Let $\mathcal{U}_{\ell,s}(\mathcal{A})$ be the finite collection of all admissible quotients of all resulting configurations, with all vertices of the root edges required to remain distinct.

These certificates force a clean set in every sufficiently dense free graph.

Lemma 3.3. Let H be a $\mathcal{U}_{\ell,s}(\mathcal{A})$ -free r -graph satisfying $\delta_\ell(H) \geq cn^s$ for some fixed $c > 0$. If n is sufficiently large, then H contains an $\mathcal{A}$ -clean ℓ -set.

Proof. First choose ℓ pairwise disjoint edges greedily. Suppose that a constant number of edges has already been chosen, and let W be their vertex union. Choose an ℓ -set T outside W . It has at least cn^s completions, whereas only $O_{r,|W|}(n^{s-1})$ completions meet W . Thus one completion gives a new edge disjoint from W .

Let $R_1, \dots, R_\ell$ be the resulting disjoint edges and let R be their union. If no $S \in \binom{R}{\ell}$ is clean, choose a failure witness for each S . A failure of (C1) gives a Type I certificate. A failure of (C2) gives a Type II

certificate, together with the support witnesses above. Map the formal vertices to the chosen host vertices and identify formal vertices exactly when their images coincide. The kernel is an admissible quotient, and its image is a member of $\mathcal{U}_{\ell,s}(\mathcal{A})$ contained in H , a contradiction. $\square$

A clean set gives the following inequalities.

Lemma 3.4. *Let H be an n -vertex r -graph, let S be $\mathcal{A}$ -clean, put $X = X_H(S)$ and $m = |X|$, and write $\delta = \delta_\ell(H)$. Then*

$$\delta \leq \binom{m}{s} \quad (3.1)$$

and

$$\delta \binom{m}{\ell} \leq \binom{n-m}{s} \text{ex}(m, \mathcal{A}). \quad (3.2)$$

If $s = \ell$, one additionally has

$$\delta \leq \text{ex}(m, \mathcal{A}). \quad (3.3)$$

Proof. Every edge of $L_H(S)$ lies in X , so $\delta \leq d_H(S) = e(L_H(S)) \leq \binom{m}{s}$, which is (3.1).

For every $T \in \binom{X}{\ell}$, one has $d_H(T) \geq \delta$. By (C1), every s -set completing T to an edge is contained in $V(H) \setminus X$; otherwise the resulting edge would contain at least $\ell + 1$ vertices of X . Hence

$$\begin{aligned} \delta \binom{m}{\ell} &\leq \sum_{T \in \binom{X}{\ell}} d_H(T) \\ &= \sum_{Q \in \binom{V(H) \setminus X}{s}} e(L_H(Q)[X]) \\ &\leq \binom{n-m}{s} \text{ex}(m, \mathcal{A}), \end{aligned}$$

where the last inequality uses (C2). This proves (3.2).

If $s = \ell$, apply (C2) with $Q = S$. Since every edge of $L_H(S)$ lies in X , the link itself is $\mathcal{A}$ -free and (3.3) follows. $\square$

Optimizing the clean-set inequalities gives the asymptotic upper bound.

Proposition 3.5. *If $1 \leq s < \ell$, then*

$$\pi_\ell^{(\ell+s)}(\mathcal{U}_{\ell,s}(\mathcal{A})) \leq \frac{\alpha}{(1 + \alpha^{1/s})^s}.$$

If $s = \ell$, then

$$\pi_\ell^{(2\ell)}(\mathcal{U}_{\ell,\ell}(\mathcal{A})) \leq \frac{\alpha}{2^\ell}.$$

Proof. It is enough to consider a sequence with normalized minimum degree bounded away from zero. Lemma 3.3 gives a clean set, and Lemma 3.4 gives the three bounds above. Put $x := m/n$ and $z := \delta / \binom{n-\ell}{s}$. Because the normalized minimum degree is bounded away from zero, (3.1) implies $m \rightarrow \infty$. After passing to a subsequence, assume x converges. Equations (3.1) and (3.2), together with $\text{ex}(m, \mathcal{A}) / \binom{m}{\ell} \rightarrow \alpha$, give

$$z \leq x^s + o(1), \quad z \leq \alpha(1-x)^s + o(1).$$

For $s < \ell$, the maximum over $x \in [0, 1]$ of the minimum of the two main terms is attained when

$$x^s = \alpha(1-x)^s, \quad x = \frac{\alpha^{1/s}}{1 + \alpha^{1/s}},$$

and equals the claimed off-diagonal bound.

If $s = \ell$, equations (3.2) and (3.3) instead give

$$z \leq \alpha x^\ell + o(1), \quad z \leq \alpha(1-x)^\ell + o(1).$$

Their balanced maximum is $\alpha/2^\ell$. $\square$

3.2 Balancing lemmas for the lower constructions

The two lower-bound constructions use randomly relabeled copies of a fixed near-extremal graph. The first lemma controls individual edges and the second controls all intersection profiles. The lower-bound constructions use randomly relabeled copies of a fixed near-extremal graph; the first lemma controls individual edges and the second controls all intersection profiles.

Lemma 3.6. *Fix integers $r \geq 1$ and $s \geq 1$. Let X_m, Y_m be disjoint sets of comparable sizes tending to infinity, and let G_m be an r -graph on X_m with*

$$p_m := \frac{e(G_m)}{\binom{|X_m|}{r}}.$$

For each $Q \in \binom{Y_m}{s}$, one can choose a relabeled copy G_Q of G_m on X_m so that every $T \in \binom{X_m}{r}$ belongs to at least

$$(p_m - o(1)) \binom{|Y_m|}{s}$$

of the graphs G_Q . The $o(1)$ is uniform in T .

Proof. Choose the relabelings independently and uniformly. For a fixed T , the indicators $\mathbf{1}_{\{T \in E(G_Q)\}}$ are independent with common mean p_m . Hoeffding's inequality [17] gives

$$\Pr \left(\sum_Q \mathbf{1}_{\{T \in E(G_Q)\}} < (p_m - \eta_m) \binom{|Y_m|}{s} \right) \leq \exp \left(-2\eta_m^2 \binom{|Y_m|}{s} \right).$$

Put $n_m := |X_m| + |Y_m|$ and $N_m := \binom{|Y_m|}{s}$. Since the two set sizes are comparable, $N_m \geq c_s n_m$ for all sufficiently large m . Taking $\eta_m := C_{r,s} \sqrt{(\log n_m)/N_m}$ with $C_{r,s}$ sufficiently large gives $\eta_m \rightarrow 0$ and makes the right side in the preceding display $o(n_m^{-r})$. A union bound over the at most $|X_m|^r \leq n_m^r$ choices of T shows that a simultaneous choice exists. $\square$

The second balancing statement controls all intersection profiles simultaneously.

Lemma 3.7. *Fix $r \geq 2$. Let X_m, Y_m be disjoint sets of comparable sizes tending to infinity, and let G_m be an r -graph on X_m with density p_m . For every $Q \in \binom{Y_m}{r}$, choose a relabeled copy G_Q of G_m on X_m , and form the $2r$ -graph*

$$H_m := \{Q \cup e : Q \in \binom{Y_m}{r}, e \in E(G_Q)\}.$$

The relabelings can be chosen so that, for every r -set R with $j := |R \cap Y_m|$,

$$d_{H_m}(R) \geq (p_m - o(1)) \binom{|Y_m| - j}{r - j} \binom{|X_m| - (r - j)}{j}. \quad (3.4)$$

The error is uniform over all R and $0 \leq j \leq r$.

Proof. Choose the relabelings independently and uniformly. Write $A = R \cap Y_m$ and $B = R \cap X_m$, so $|A| = j$ and $|B| = r - j$. For every $Q \supseteq A$, put $Z_Q := d_{G_Q}(B)$. These variables are independent and satisfy $0 \leq Z_Q \leq M_j := \binom{|X_m| - (r - j)}{r - j}$. By symmetry, $\mathbb{E} Z_Q = p_m M_j$. If $j = r$, then there is one possible $Q = R$ and $Z_Q = e(G_Q) = p_m \binom{|X_m|}{r}$, so the assertion is deterministic. If $j < r$, normalize by M_j and apply Hoeffding. The number of independent summands is $N_j := \binom{|Y_m| - j}{r - j} \geq c_r |Y_m|$ for all sufficiently large m . Taking $\eta_m = C_r \sqrt{\log(|X_m| + |Y_m|)/|Y_m|}$ with C_r sufficiently large, the probability that the sum is below $(p_m - \eta_m) N_j M_j$ is $o((|X_m| + |Y_m|)^{-r})$. A union bound over all r -sets R proves the claim. $\square$

3.3 Lower constructions

We first treat the off-diagonal case $1 \leq s < \ell$. Let

$$x := \frac{\alpha^{1/s}}{1 + \alpha^{1/s}}, \quad y := 1 - x.$$

Partition an n -vertex set as $V = X \dot{\cup} Y$, where $|X| = xn + o(n)$ and $|Y| = yn + o(n)$. Choose an $\mathcal{A}$ -free ℓ -graph G on X with $e(G)/\binom{|X|}{\ell} = \alpha - o(1)$. For every $Q \in \binom{Y}{s}$, use Lemma 3.6 to choose a relabeled copy G_Q of G on X so that every $T \in \binom{X}{\ell}$ belongs to $(\alpha - o(1))\binom{|Y|}{s}$ of the G_Q . Define an r -graph H_n by the following two edge families:

(L1) $T \cup Q$ whenever $Q \in \binom{Y}{s}$ and $T \in E(G_Q)$;

(L2) every r -set E satisfying

$$s + 1 \leq |E \cap Y| \leq \ell.$$

The second family consists of the *guard edges*.

We first verify the minimum degree of this construction.

Lemma 3.8. *The graph H_n is $\mathcal{U}_{\ell,s}(\mathcal{A})$ -free and satisfies*

$$\frac{\delta_\ell(H_n)}{\binom{n-\ell}{s}} \geq \frac{\alpha}{(1 + \alpha^{1/s})^s} - o(1).$$

Proof. Let R be an ℓ -set and put $j = |R \cap Y|$. If $j = 0$, incidence balancing of the copies of G gives

$$d_{H_n}(R) \geq (\alpha - o(1)) \binom{|Y|}{s},$$

whose normalized limit is αy^s . If $j = \ell$, every s -set of X completes R to a guard edge, so the normalized limit is x^s .

Suppose $1 \leq j \leq \ell - 1$. Choose

$$h := \begin{cases} s + 1 - j, & j \leq s, \\ 0, & j \geq s + 1. \end{cases}$$

Then $0 \leq h \leq s$ and $s + 1 \leq j + h \leq \ell$. Complete R by choosing h vertices from Y and $s - h$ from X . Every such completion is a guard edge. The normalized contribution of these completions is

$$\binom{s}{h} y^h x^{s-h} \geq \min\{x^s, y^s\} \geq \min\{x^s, \alpha y^s\}.$$

Here $\alpha \leq 1$ because it is an ordinary Turán density. By the choice of x and y , $x^s = \alpha y^s$, which proves the claimed bound.

It remains to prove freeness. Suppose a forbidden configuration is embedded. Its root part gives ℓ pairwise disjoint edges $R_1, \dots, R_\ell$. Every edge of H_n contains at least one vertex of Y , so choose $y_i \in R_i \cap Y$ and put $S := \{y_1, \dots, y_\ell\} \subseteq Y$. The root edges are disjoint, so S is an ℓ -set in their union.

Every edge containing S must have exactly ℓ vertices in Y , and all such edges are guard edges. Therefore

$$L_{H_n}(S) = \binom{X}{s}, \quad X_{H_n}(S) = X.$$

Every edge copied from G has exactly ℓ vertices in X , and every guard edge has at most $\ell - 1$ vertices in X . Thus (C1) holds.

For $Q \in \binom{V(H_n)}{s}$, if $Q \subseteq Y$, then $L_{H_n}(Q)[X] = G_Q$, which is $\mathcal{A}$ -free. If $Q \cap X \neq \emptyset$, then $L_{H_n}(Q)[X]$ is empty, because an edge formed with an ℓ -set of X would contain more than ℓ vertices of X . Hence (C2) also holds, so S is clean. This contradicts the certificate attached to S in the forbidden configuration. $\square$

The diagonal case $s = \ell$. When $s = \ell$, the guard range above is empty, so the preceding lower construction cannot simply be specialized to the diagonal. Instead, exact-profile balancing gives the degree bound simultaneously for every intersection profile $j = |R \cap Y|$. The avoidance argument still uses the same clean-set certificate system.

Partition $V = X \dot{\cup} Y$ with $|X|, |Y| = n/2 + o(n)$. Choose an $\mathcal{A}$ -free ℓ -graph G on X of density $\alpha - o(1)$. Apply Lemma 3.7 and define the 2ℓ -graph

$$H_n := \{Q \cup e : Q \in \binom{Y}{\ell}, e \in E(G_Q)\}.$$

Exact-profile balancing gives the degree estimate; freeness follows from the clean-set witness exactly as in the off-diagonal construction.

Lemma 3.9. *The graphs H_n are $\mathcal{U}_{\ell,\ell}(\mathcal{A})$ -free and satisfy*

$$\frac{\delta_\ell(H_n)}{\binom{n-\ell}{\ell}} \geq \frac{\alpha}{2^\ell} - o(1).$$

Proof. For an ℓ -set R with $j = |R \cap Y|$, equation (3.4) gives

$$\frac{d_{H_n}(R)}{\binom{n-\ell}{\ell}} \geq (\alpha - o(1)) \binom{\ell}{j} 2^{-\ell} \geq \frac{\alpha}{2^\ell} - o(1).$$

For avoidance, suppose that a forbidden configuration is embedded. Choose one Y -vertex from each of its ℓ disjoint root edges and call the resulting ℓ -set $S \subseteq Y$. Then $L_{H_n}(S) = G_S$ and $X_{H_n}(S) \subseteq X$. Every edge has exactly ℓ vertices in X , so (C1) holds. If $Q \subseteq Y$, then $L_{H_n}(Q)[X_{H_n}(S)]$ is an induced subgraph of the $\mathcal{A}$ -free graph G_Q ; if Q meets X , the restricted link is empty. Thus S is clean, contradicting its attached certificate. $\square$

Combining the preceding results gives the first exact transfer theorem.

Theorem 3.10. *Let $\ell \geq 2$, $1 \leq s \leq \ell$, and let $\mathcal{A}$ be a finite family of ℓ -graphs with ordinary Turán density $\alpha > 0$. The finite family $\mathcal{U}_{\ell,s}(\mathcal{A})$ is effectively computable from the edge lists of $\mathcal{A}$ and satisfies*

$$\pi_\ell^{(\ell+s)}(\mathcal{U}_{\ell,s}(\mathcal{A})) = \begin{cases} \frac{\alpha}{(1 + \alpha^{1/s})^s}, & 1 \leq s < \ell, \\ \frac{\alpha}{2^\ell}, & s = \ell. \end{cases}$$

The algorithm producing the forbidden family from the edge lists of $\mathcal{A}$ does not use the numerical value of α ; that value enters only the density analysis and the asymptotic lower-bound construction.

Proof. The upper bounds are Proposition 3.5. For $s < \ell$, the matching lower bound follows from Lemma 3.8. For $s = \ell$, it follows from Lemma 3.9. The family is finite and effective because there are finitely many certificate assignments and finitely many admissible quotients of each finite formal configuration. $\square$

At the codegree endpoint, this transfer gives the familiar ordinary-to-codegree formula.

Remark 3.11. *For $s = 1$, the formula in Theorem 3.10 becomes $\alpha/(1 + \alpha)$. The endpoint limit theory was initiated by Mubayi and Zhao [32]. The clean-set theorem proves this formula together with its extension through the entire range $1 \leq s < \ell$.*

4 Marked suspension

The construction for large s requires ordinary families with near-extremal examples whose marker part is determined by the edges. The section proceeds in three steps: we give the suspension class a finite ordinary basis, compute its exact Lagrangian, and then prove the one-marker realizability needed by the two-marker transfer.

4.1 The suspension class

Fix integers $2 \leq q < s$ and put $t := s - q$. Let $\mathcal{D} = \text{Forb}(\mathcal{A})$ be a finitely based monotone class of q -graphs. Define the t -fold suspension class $\mathcal{S}_t(\mathcal{D})$ to consist of all edgeless s -graphs and all nonempty s -graphs H for which there is a t -set R , called a *common root*, contained in every edge and $L_H(R) \in \mathcal{D}$.

We first verify that the suspension class has a finite ordinary description.

Proposition 4.1. *The class $\mathcal{S}_t(\mathcal{D})$ is monotone and has an effectively computable finite ordinary forbidden basis.*

Proof. Monotonicity is immediate: a nonempty subgraph retains the same common root and its root link is a subgraph of the old link; edgeless subgraphs are included by definition.

Let $H \notin \mathcal{S}_t(\mathcal{D})$. Then H has an edge. Fix $E_0 \in E(H)$ and put $I := \bigcap_{E \in E(H)} E$. For every $v \in E_0 \setminus I$, choose an edge E_v missing v . The intersection of these at most $s+1$ selected edges is exactly I . If $|I| < t$, the subgraph consisting of these edges is already a nonmember on at most $s(s+1)$ vertices.

Suppose $|I| \geq t$. For every t -set $R \subseteq I$, the link $L_H(R)$ does not belong to $\mathcal{D}$, and hence contains a member $A_R \in \mathcal{A}$; in particular, this case forces $\mathcal{A} \neq \emptyset$. Let K contain the selected edges, and for every such R include all vertices of one injective copy of A_R in $L_H(R)$, including its isolated vertices, together with the edges $R \cup e$ for $e \in E(A_R)$. If $v_{\max} := \max_{A \in \mathcal{A}} v(A)$, then

$$v(K) \leq s(s+1) + \binom{s}{t} v_{\max}.$$

Every common t -set of the edges of K lies in I ; for that particular root, its link in K contains the selected copy of A_R . Hence $K \notin \mathcal{S}_t(\mathcal{D})$.

Membership in $\mathcal{S}_t(\mathcal{D})$ is decidable: enumerate the common t -sets and test their finite links for copies of members of $\mathcal{A}$. Enumerating all nonmembers up to the displayed order bound therefore gives an effective finite forbidden basis. Monotonicity shows that every nonmember contains one of the enumerated obstructions. $\square$

Its Lagrangian follows from a direct optimization of the root and link weights.

Proposition 4.2. *One has*

$$\Lambda(\mathcal{S}_t(\mathcal{D})) = \frac{q^q}{s^s} \Lambda(\mathcal{D}). \quad (4.1)$$

Consequently, applying Theorem 2.4 to the finite basis from Proposition 4.1 gives a finite family $\mathcal{A}_s(\mathcal{D})$ of s -graphs satisfying

$$\pi(\mathcal{A}_s(\mathcal{D})) = \kappa_{q,s} \beta(\mathcal{D}), \quad \kappa_{q,s} := \frac{s!q^q}{q!s^s}, \quad \beta(\mathcal{D}) := q! \Lambda(\mathcal{D}). \quad (4.2)$$

Proof. Let $H \in \mathcal{S}_t(\mathcal{D})$ be nonempty with common root R , and let x be a probability weighting on $V(H)$. Put $z := \sum_{v \notin R} x_v$. Then AM–GM on the t root weights gives

$$p_H(x) \leq \left(\frac{1-z}{t} \right)^t \Lambda(\mathcal{D}) z^q.$$

The right side is maximized at $z = q/s$ and equals $\Lambda(\mathcal{D}) q^q / s^s$. Conversely, take increasingly good weighted members of $\mathcal{D}$, adjoin t common root vertices, put weight $1/s$ on each root, and scale the data weights to total mass q/s . This approaches the upper bound and proves (4.1). Equation (4.2) follows from Theorem 2.4. $\square$

4.2 One-marker realizability

For the later two-marker construction, we need near-extremal examples with a distinguished marker part.

Definition 4.3. *Let $\mathcal{A}$ be a finite family of s -graphs with ordinary density a . We say that $\mathcal{A}$ is one-marker realizable if, for every sufficiently large integer n , there is an n -vertex $\mathcal{A}$ -free s -graph G_n with a marker partition $V(G_n) = V_0 \dot{\cup} V_1$ such that, as $n \rightarrow \infty$,*

$$\frac{|V_0|}{n} = \frac{1}{s} + o(1),$$

every edge contains exactly one vertex of V_0 , and

$$\frac{e(G_n)}{\binom{n}{s}} = a - o(1).$$

The errors may depend on $\mathcal{A}$ and s but are uniform over all sufficiently large n .

Suspensions provide such marker realizations automatically.

Lemma 4.4. *For every finitely based $\mathcal{D}$ as above, the family $\mathcal{A}_s(\mathcal{D})$ from Proposition 4.2 is one-marker realizable.*

Proof. Choose finite weighted members $D_j \in \mathcal{D}$ whose Lagrangians tend to $\Lambda(\mathcal{D})$, adjoin common roots $z_1, \dots, z_t$, and put mass $1/s$ on every root and total mass q/s on the data link. Let m_j be the number of weighted classes in the resulting j th weighted graph. Choose a nondecreasing function $j(n) \rightarrow \infty$ so slowly that $m_{j(n)}/n \rightarrow 0$; for example, after choosing integers $N_j \geq jm_j$ increasing in j , take $j(n) := \max\{j : N_j \leq n\}$. For target order n , use this weighted graph with index $j(n)$ and round all class sizes to integers summing to n . If x is the target weight vector and y the rounded vector, then $\|x - y\|_1 = O(m_{j(n)}/n) = o(1)$. For every s -graph J and probability vectors x, y on its vertex set,

$$|p_J(x) - p_J(y)| \leq \frac{\|x - y\|_1}{(s-1)!}.$$

Indeed, every coordinate derivative of p_J is at most $e_{s-1}(w) \leq 1/(s-1)!$ at every probability vector w , and integration along the segment from x to y gives the claim. Thus the rounding error is uniform even though the approximating graph varies with n . It also gives each root class size $n/s + o(n)$, and the resulting blowup has density tending to (4.2) for every sufficiently large n .

Every edge uses exactly one vertex from each root class. Designate the first root class as V_0 and merge the other root classes and all data classes into V_1 . Then every edge contains exactly one V_0 -vertex and $|V_0|/n = 1/s + o(1)$. The blowups are $\mathcal{A}_s(\mathcal{D})$ -free by the lower-bound argument in Theorem 2.4. If isolated vertices were deleted from the finite forbidden family, Lemma 2.2 shows that the same conclusion holds for all sufficiently large n . $\square$

Let $\mathcal{D}_\top$ denote the class of all finite q -graphs. It is finitely based by the empty forbidden family and satisfies $\beta(\mathcal{D}_\top) = 1$. Define the fixed *reference family* $\mathcal{A}_s^* := \mathcal{A}_s(\mathcal{D}_\top)$. Then $\pi(\mathcal{A}_s^*) = \kappa_{q,s}$, and $\mathcal{A}_s^*$ is one-marker realizable.

For the guard calculation in the next section, put $\gamma_s := ((s-1)/s)^s$.

Lemma 4.5. *For every fixed $q \geq 2$, there is an effectively computable integer $s_0(q) > q$ such that*

$$\kappa_{q,s} \leq \gamma_s \quad \text{for every } s \geq s_0(q).$$

Proof. The sequence $\gamma_s = (1 - 1/s)^s$ is increasing for $s \geq 2$ and starts at $1/4$. Indeed, the function $x \mapsto x \log(1 - 1/x)$ has positive derivative for $x > 1$, since $u - \log(1 + u) > 0$ for $u = 1/(x-1) > 0$. Also

$$\kappa_{q,s} = \frac{q^q}{q!} \frac{s!}{s^s} \leq \frac{q^q}{q!} \left(\frac{s+1}{2s} \right)^s \leq \frac{q^q}{q!} \left(\frac{3}{4} \right)^s,$$

where the first inequality is AM–GM applied to $1/s, 2/s, \dots, s/s$. Moreover,

$$\frac{\kappa_{q,s+1}}{\kappa_{q,s}} = \left(\frac{s}{s+1} \right)^s < 1,$$

whereas γ_s is increasing. Thus once $\kappa_{q,s} \leq \gamma_s$ holds, it holds for every larger s . Equivalently, one may search for the first $S > q$ satisfying the explicit condition $(q^q/q!)(3/4)^S \leq 1/4$. For every $s \geq S$, the left side only decreases and $\gamma_s \geq 1/4$, so $s_0(q) := S$ is an exactly computable valid choice. $\square$

5 The two-marker transfer

Fix $\ell \geq 2$ and $s \geq 2$, and put $r = \ell + s$. Let $\mathcal{A}$ and $\mathcal{A}^*$ be finite families of s -graphs. Normalize them by Lemma 2.2. This normalization preserves one-marker realizability, because the original and normalized families have exactly the same free hosts for every sufficiently large order. Write $a := \pi(\mathcal{A})$ and $b := \pi(\mathcal{A}^*)$. Assume

$$0 < a \leq b \leq \gamma_s$$

and that both families are one-marker realizable.

The assumptions are used at distinct points. Positivity of a places the argument on the positive n^s degree scale. The ordering $a \leq b$ fixes the orientation of the optimizing pair below and gives $x \geq y$. The bound

$b \leq \gamma_s$ is used only in the mixed-profile guard estimate. One-marker realizability is needed only for the four-part lower construction; neither the clean-pair certificate nor the upper bound uses it. Finally, $\ell \geq 2$ makes the three lower-construction edge families disjoint, and $s \geq 2$ ensures that every mixed ℓ -set has the guard completions used below. The certificate family itself is computed solely from the two input edge lists, without knowing a or b .

5.1 Clean pairs and the upper bound

For disjoint ℓ -sets S, T in an r -graph H , put $X := X_H(S)$ and $Y := X_H(T)$.

Definition 5.1. *The ordered pair (S, T) is $(\mathcal{A}, \mathcal{A}^*)$ -clean if*

- (P1) $X \cap Y = \emptyset$;
- (P2) $L_H(S)[X]$ is $\mathcal{A}$ -free;
- (P3) $L_H(T)[Y]$ is $\mathcal{A}^*$ -free.

Each failure has a bounded explicit certificate.

Overlap certificate. Choose s -sets P, Q and a marked vertex $z \in P \cap Q$, and prescribe

$$S \cup P, \quad T \cup Q.$$

$\mathcal{A}$ -link certificate. Choose $A \in \mathcal{A}$, an injective formal copy U of A disjoint from S , and prescribe

$$S \cup e \quad (e \in E(A)).$$

Since A has no isolated vertex, every copy vertex lies in a prescribed edge and is therefore certified to belong to $X_H(S)$. Define the $\mathcal{A}^*$ -link certificate analogously, rooted at T .

Take ℓ pairwise disjoint formal r -edges $R_1, \dots, R_\ell$ and let R be their union. For every ordered pair of disjoint ℓ -sets

$$S, T \in \binom{R}{\ell}, \quad S \cap T = \emptyset,$$

choose one of the three certificate types above. Let $\mathcal{B}_{\ell,s}(\mathcal{A}, \mathcal{A}^*)$ be the finite collection of all admissible quotients in the sense of Definition 3.2, with the root-edge vertices required to remain distinct.

These certificates force a clean pair under an n^s -scale lower bound on $\delta_\ell(H)$.

Lemma 5.2. *Let H be a $\mathcal{B}_{\ell,s}(\mathcal{A}, \mathcal{A}^*)$ -free r -graph with $\delta_\ell(H) \geq cn^s$ for some fixed $c > 0$. For sufficiently large n , the graph H contains an $(\mathcal{A}, \mathcal{A}^*)$ -clean ordered pair of disjoint ℓ -sets.*

Proof. As in Lemma 3.3, the bound $\delta_\ell(H) \geq cn^s$ gives ℓ pairwise disjoint edges $R_1, \dots, R_\ell$. If every ordered disjoint pair (S, T) in their union failed cleanliness, choose an actual certificate of one of the three types. Identifying formal vertices precisely when their images coincide produces an admissible quotient contained in H , a contradiction. $\square$

A clean pair bounds the minimum degree through its two disjoint link supports.

Proposition 5.3. *One has*

$$\pi_\ell^{(\ell+s)}(\mathcal{B}_{\ell,s}(\mathcal{A}, \mathcal{A}^*)) \leq \Omega_s(a, b) := \frac{ab}{(a^{1/s} + b^{1/s})^s}.$$

Proof. Suppose for a contradiction that the claimed upper bound fails. Choose a sequence of $\mathcal{B}_{\ell,s}(\mathcal{A}, \mathcal{A}^*)$ -free n -vertex r -graphs whose normalized minimum degrees converge to a number $d > \Omega_s(a, b) > 0$. For some fixed $c > 0$, all sufficiently large members of this sequence satisfy $\delta_\ell(H) \geq cn^s$, so Lemma 5.2 gives a clean pair (S, T) . Put $X = X_H(S)$, $Y = X_H(T)$, $m = |X|$, $p = |Y|$, and $\delta := \delta_\ell(H)$. Every edge of $L_H(S)$ lies in X , and the restricted link is $\mathcal{A}$ -free. Hence

$$\delta \leq d_H(S) = e(L_H(S)) \leq \text{ex}(m, \mathcal{A}).$$

Similarly,

$$\delta \leq \text{ex}(p, \mathcal{A}^*).$$

Because the normalized minimum degree is positive, these two bounds imply $m, p \rightarrow \infty$. Cleanliness gives $m + p \leq n$. Writing $x = m/n$, $y = p/n$, and normalizing by $\binom{n-\ell}{s}$, every subsequential limit d satisfies

$$d \leq ax^s, \quad d \leq by^s, \quad x + y \leq 1.$$

The maximum of $\min\{ax^s, by^s\}$ subject to $x + y \leq 1$ occurs when $x + y = 1$ and $ax^s = by^s$. Thus

$$x = \frac{b^{1/s}}{a^{1/s} + b^{1/s}}, \quad y = \frac{a^{1/s}}{a^{1/s} + b^{1/s}}, \quad (5.1)$$

and the common value is $\Omega_s(a, b)$. $\square$

5.2 The four-part lower construction

Take x, y from (5.1). Since $a \leq b$, one has $x \geq y$. First partition an n -vertex set as $X \dot{\cup} Y$ with $|X| = xn + o(n)$ and $|Y| = yn + o(n)$. Choose one-marker realizations on X and Y . Let $X = X_0 \dot{\cup} X_1$ and $Y = Y_0 \dot{\cup} Y_1$ be the marker partitions given by the corresponding realizations. Then

$$\begin{aligned} \frac{|X_0|}{n} &= \frac{x}{s} + o(1), & \frac{|X_1|}{n} &= \frac{(s-1)x}{s} + o(1), \\ \frac{|Y_0|}{n} &= \frac{y}{s} + o(1), & \frac{|Y_1|}{n} &= \frac{(s-1)y}{s} + o(1). \end{aligned}$$

We have

- an $\mathcal{A}$ -free s -graph G_X on X , every edge of which contains exactly one vertex of X_0 , with density $a - o(1)$;
- an $\mathcal{A}^*$ -free s -graph G_Y on Y , every edge of which contains exactly one vertex of Y_0 , with density $b - o(1)$.

Define an r -graph H_n by three edge families:

(M1) $S \cup e$ for every $S \in \binom{Y_0}{\ell}$ and $e \in E(G_X)$;

(M2) $T \cup f$ for every $T \in \binom{X_0}{\ell}$ and $f \in E(G_Y)$;

(M3) every r -set E satisfying

$$1 \leq |E \cap X_0| \leq \ell - 1, \quad 1 \leq |E \cap Y_0| \leq \ell - 1.$$

The third family consists of the *guard edges*. The three families are disjoint because $\ell \geq 2$ and the first two have marker profiles $(1, \ell)$ and $(\ell, 1)$, respectively.

We first compute the minimum degree of the four-part construction.

Lemma 5.4. *The graph H_n is $\mathcal{B}_{\ell, s}(\mathcal{A}, \mathcal{A}^*)$ -free and satisfies*

$$\frac{\delta_\ell(H_n)}{\binom{n-\ell}{s}} \geq \Omega_s(a, b) - o(1).$$

Proof. If $R \in \binom{Y_0}{\ell}$, then $L_{H_n}(R) = G_X$, so the normalized degree is at least $ax^s - o(1)$. If $R \in \binom{X_0}{\ell}$, the analogous value is $by^s - o(1)$. By (5.1), both equal $\Omega_s(a, b)$ in the limit.

Now suppose that R is neither contained in X_0 nor in Y_0 . Put $u := |R \cap X_0|$ and $v := |R \cap Y_0|$, so $0 \leq u, v \leq \ell - 1$. Let $i := \mathbf{1}_{\{u=0\}}$ and $j := \mathbf{1}_{\{v=0\}}$. Complete R by choosing i vertices from X_0 , j vertices from Y_0 , and $s - i - j$ vertices from X_1 . Since $s \geq 2$, this is valid, and every such completion is a guard edge. The four marker profiles and their normalized contributions are

(i, j)	asymptotic normalized main term
$(0, 0)$	$\gamma_s x^s$
$(1, 0)$	$x \left(\frac{(s-1)x}{s} \right)^{s-1}$
$(0, 1)$	$y \left(\frac{(s-1)y}{s} \right)^{s-1}$
$(1, 1)$	$\frac{s-1}{s} xy \left(\frac{(s-1)x}{s} \right)^{s-2}$

Because $x \geq y$, each of these four expressions is at least

$$\gamma_s y x^{s-1}.$$

On the other hand, $x/y = (b/a)^{1/s}$, and hence

$$ax^s = a^{(s-1)/s} b^{1/s} y x^{s-1}.$$

Since $a \leq b \leq \gamma_s$, we have $a^{(s-1)/s} b^{1/s} \leq b \leq \gamma_s$. Thus the guard contribution is at least $ax^s = \Omega_s(a, b)$, which proves the degree bound.

It remains to prove freeness. Suppose that a forbidden configuration is embedded, with pairwise disjoint root edges $R_1, \dots, R_\ell$. Every edge of H_n contains at least one vertex of X_0 and at least one vertex of Y_0 . Choose $x_i \in R_i \cap X_0$ and $y_i \in R_i \cap Y_0$, and put $T := \{x_1, \dots, x_\ell\} \subseteq X_0$ and $S := \{y_1, \dots, y_\ell\} \subseteq Y_0$. These are disjoint ℓ -sets in the root union.

The marker profiles imply

$$L_{H_n}(S) = G_X, \quad L_{H_n}(T) = G_Y.$$

Their supports lie in the disjoint sets X and Y , respectively. The first link is $\mathcal{A}$ -free and the second is $\mathcal{A}^*$ -free. Hence (S, T) is clean, contradicting the failure certificate attached to this ordered pair in the forbidden configuration. $\square$

The upper bound and the two lower-construction lemmas now give the exact transfer.

Theorem 5.5. *Let $\ell, s \geq 2$. Let $\mathcal{A}, \mathcal{A}^*$ be finite families of s -graphs that are one-marker realizable and have ordinary densities a, b satisfying $0 < a \leq b \leq \gamma_s$. Then the finite family $\mathcal{B}_{\ell, s}(\mathcal{A}, \mathcal{A}^*)$ is effectively computable and*

$$\pi_\ell^{(\ell+s)}(\mathcal{B}_{\ell, s}(\mathcal{A}, \mathcal{A}^*)) = \frac{ab}{(a^{1/s} + b^{1/s})^s}.$$

The algorithm producing the forbidden family from the two edge lists does not use the numerical values a, b ; those values enter only the density analysis and the asymptotic lower-bound construction.

Proof. The upper bound is [Proposition 5.3](#). The lower bound follows from [Lemma 5.4](#). Finiteness and effectivity follow from the bounded certificate system and finite quotient enumeration. $\square$

We finish the section by combining [Theorem 5.5](#) with marked suspension. Let $\mathcal{D}$ be a finitely based q -graph class satisfying $\Lambda(\mathcal{D}) > 0$, and put $\beta := q! \Lambda(\mathcal{D})$. Then $0 < \beta \leq 1$: for every q -graph G and probability vector z ,

$$q! p_G(z) \leq q! e_q(z) \leq \left(\sum_{v \in V(G)} z_v \right)^q = 1.$$

For $s \geq s_0(q)$, take $\mathcal{A} = \mathcal{A}_s(\mathcal{D})$ and $\mathcal{A}^* = \mathcal{A}_s^*$. Then $a = \kappa_{q, s} \beta$ and $b = \kappa_{q, s}$. By [Lemmas 4.4](#) and [4.5](#), the hypotheses of the two-marker theorem hold. We obtain a finite family $\mathcal{M}_{q, s, \ell}(\mathcal{D})$ of $(\ell + s)$ -graphs satisfying

$$\pi_\ell^{(\ell+s)}(\mathcal{M}_{q, s, \ell}(\mathcal{D})) = \Gamma_{q, s}(\beta), \tag{5.2}$$

where

$$\Gamma_{q, s}(z) := \frac{\kappa_{q, s} z}{(1 + z^{1/s})^s}.$$

The arithmetic behavior of the two transfer maps is summarized next.

Lemma 5.6. *For fixed positive integers s, q , the maps*

$$\Theta_s(z) := \frac{z}{(1 + z^{1/s})^s}, \quad \Gamma_{q, s}(z) := \kappa_{q, s} \Theta_s(z)$$

are continuous, strictly increasing, and semialgebraic over $\mathbb{Q}$ on $[0, 1]$. On $(0, 1]$ they have algebraic inverse formulas. In particular, if z is transcendental, then so are $\Theta_s(z)$ and $\Gamma_{q, s}(z)$.

Proof. Writing $u = z^{1/s}$ gives

$$\Theta_s(z)^{1/s} = \frac{u}{1+u},$$

which is strictly increasing. If $w = \Theta_s(z)^{1/s}$, then

$$z = \left(\frac{w}{1-w} \right)^s.$$

For $\Gamma_{q,s}$, replace w by $(\Gamma_{q,s}(z)/\kappa_{q,s})^{1/s}$. The maps are semialgebraic because s -th roots are described by polynomial equations with a nonnegativity condition. If an output were algebraic, the inverse formula would make the input algebraic. $\square$

6 Ordinary input constructions

We need two ordinary inputs: a fixed-order score construction for undecidability and a finite-family transcendence construction. We prove the first and state the precise form of the second that is used later.

6.1 A fixed-order construction for undecidability

Let $\mathcal{U}_d$ be the set of isomorphism classes of simple graphs on d vertices. If (G, x) is a finite weighted graph, sample d vertices independently according to x and let $G_d(G, x)$ be the graph induced on the ordered sample, with repeated samples represented by repeated indices. For $F \in \mathcal{U}_d$, define the evaluation of the *induced-density basis element* $I_d(F)$ by

$$t(I_d(F); G, x) := \Pr([G_d(G, x)] = F).$$

Thus $\mathbf{1}_d := \sum_{F \in \mathcal{U}_d} I_d(F)$ evaluates to 1. For

$$f = \sum_{F \in \mathcal{U}_d} a_F I_d(F), \quad 0 \leq a_F \leq 1,$$

define the *value* of f by

$$\text{Val}_d(f) := \sup_{A, x} t(f; A, x).$$

If G is a finite graph with vertex weights x , define the *injective form*

$$T_f(G, x) := \sum_{X \in \binom{V(G)}{d}} a_{[G[X]]} \prod_{v \in X} x_v.$$

This induced-sampling language is standard in dense graph limits, as developed by Lovász and Szegedy [26] and in Lovász's monograph [25]; it is also closely related to Razborov's flag-algebra formalism [34].

The following lemma is the standard passage between weighted sampling, injective densities, and uniform blowups; see Lovász [25, Sections 5.2.2–5.2.3].

Lemma 6.1. *If every $a_F \geq 0$, then*

$$\text{Val}_d(f) = d! \sup_{G, x} T_f(G, x).$$

Moreover, there are finite graphs G_m with uniform vertex weights and $v(G_m) \rightarrow \infty$ such that

$$d! T_f(G_m, \text{unif}) \rightarrow \text{Val}_d(f).$$

We next give the fixed-order construction used in the undecidability reduction. The proof combines a fixed-variable Diophantine representation with the ordinary graph localization of Hatami and Norine [16]. We include the reduction because the fixed sample order and the exact equality level $1/2$ are both essential below.

A *rational quantum graph* is a finite rational linear combination of finite simple graphs. If J is a graph and (G, x) is a finite weighted graph, put

$$t(J; G, x) := \sum_{\phi: V(J) \rightarrow V(G)} \prod_{u \in V(J)} x_{\phi(u)} \prod_{uv \in E(J)} \mathbf{1}_{\{\phi(u)\phi(v) \in E(G)\}},$$

and extend this evaluation linearly. We also use partially labeled quantum graphs. Their product takes disjoint union and identifies equally labeled vertices. If the label set is $[k]$ and $\phi: [k] \rightarrow V(G)$, define the rooted evaluation by

$$t(J; G, x, \phi) := \sum_{\substack{\psi: V(J) \rightarrow V(G) \\ \psi|_{[k]} = \phi}} \prod_{u \in V(J) \setminus [k]} x_{\psi(u)} \prod_{uv \in E(J)} \mathbf{1}_{\{\psi(u)\psi(v) \in E(G)\}},$$

and again extend linearly. For a partially labeled graph J , Boolean Möbius inversion gives

$$\text{ind}(J) := \sum_{\substack{J' \supseteq J \\ V(J') = V(J)}} (-1)^{e(J') - e(J)} J'.$$

Thus, after the labeled vertices have been mapped into a host, $\text{ind}(J)$ is the indicator that all prescribed adjacencies and non-adjacencies of J are respected. We write $\langle\langle f \rangle\rangle$ for the quantum graph obtained by forgetting all labels. With the convention above,

$$t(\langle\langle f \rangle\rangle; G, x) = \sum_{\phi: [k] \rightarrow V(G)} \left(\prod_{i=1}^k x_{\phi(i)} \right) t(f; G, x, \phi).$$

We recall the edge–triangle region used by Hatami and Norine. Put $b(x) := 2x^2 - x$. For $m \geq 1$ and $1 - 1/m \leq x \leq 1 - 1/(m+1)$, let

$$L(x) := \frac{3m^2 - m - 2}{m(m+1)}x - \frac{2(m-1)}{m+1},$$

and put $L(1) := 1$ and

$$\mathcal{R}_{\text{et}} := \{(x, y) \in [0, 1]^2 : y \geq L(x)\}.$$

The polynomial b is convex, and each linear piece of L is the chord joining two consecutive clique-grid points of b ; hence $L(x) \geq b(x)$ on $[0, 1]$. Bollobás’s edge–triangle inequality [5] implies that every pair $(t(K_2; G, x), t(K_3; G, x))$ from a finite weighted graph belongs to $\mathcal{R}_{\text{et}}$; equivalently, this is the normalization in Hatami and Norine [16, Theorem 5.2]. The weighted statement follows from rational blowups and continuity. The uniform clique K_n gives the boundary point

$$\left(1 - \frac{1}{n}, b\left(1 - \frac{1}{n}\right)\right).$$

The following fixed-degree version of the Hatami–Norine localization is the point at which several independent edge–triangle variables are encoded inside one ordinary graph.

Lemma 6.2 (Fixed-order localization). *Fix $k \geq 6$ and $\Delta \geq 2$. There is an effective linear map $q \mapsto \mathbf{L}(q)$ from integer polynomials*

$$q = q(x_1, \dots, x_k, y_1, \dots, y_k), \quad \deg q \leq \Delta,$$

to integer quantum graphs all of whose constituents have exactly

$$d(k, \Delta) := k(3\Delta + 1)$$

vertices, with the following properties.

- (i) *If $q \geq 0$ on $\mathcal{R}_{\text{et}}^k$, then $t(\mathbf{L}(q); G, x) \geq 0$ for every finite weighted graph (G, x) .*
- (ii) *If positive integers $n_1, \dots, n_k$ satisfy*

$$q\left(1 - \frac{1}{n_1}, \dots, 1 - \frac{1}{n_k}, b\left(1 - \frac{1}{n_1}\right), \dots, b\left(1 - \frac{1}{n_k}\right)\right) < 0,$$

then $t(\mathbf{L}(q); G) < 0$ for an effectively constructible finite graph G .

(iii) *The evaluation of $\mathbb{L}(q)$ on the one-vertex edgeless graph is zero.*

Proof. Fix an effectively constructible stringent graph H on the labeled vertex set $[k]$; such a graph exists for every $k \geq 6$ by Hatami and Norine [16, Lemma 4.2]. Here stringent means that H has no nontrivial automorphism and no nontrivial homogeneous set, where $W \subseteq V(H)$ is homogeneous when $N_H(u) \setminus W = N_H(v) \setminus W$ for all $u, v \in W$.

For $i \in [k]$ and $m \in \{1, 2, 3\}$, add to H new unlabeled vertices $z_1, \dots, z_m$ that form a clique and, for each $j \neq i$, join every z_s to j exactly when $ij \in E(H)$. Sum $\text{ind}(J)$ over all 2^m choices of the edges iz_s . Denote the resulting $[k]$ -labeled quantum graphs by V_i, E_i, T_i for $m = 1, 2, 3$, respectively. Let ψ_H be the algebra homomorphism defined by

$$\psi_H(v_i) = V_i, \quad \psi_H(e_i) = E_i, \quad \psi_H(t_i) = T_i.$$

For a polynomial q of degree at most Δ , set

$$\tau_\Delta(q) := \left(\prod_{i=1}^k v_i^{3\Delta} \right) q \left(\frac{e_1}{v_1^2}, \dots, \frac{e_k}{v_k^2}, \frac{t_1}{v_1^3}, \dots, \frac{t_k}{v_k^3} \right)$$

and

$$\mathbb{L}(q) := \langle \langle \psi_H(\tau_\Delta(q)) \rangle \rangle.$$

This is an effective integer quantum graph. Indeed, a monomial $\prod_i x_i^{\alpha_i} y_i^{\beta_i}$ is sent, before applying ψ_H , to

$$\prod_{i=1}^k v_i^{3\Delta - 2\alpha_i - 3\beta_i} e_i^{\alpha_i} t_i^{\beta_i}.$$

All exponents are nonnegative because $2\alpha_i + 3\beta_i \leq 3\Delta$. For each i , the corresponding product introduces exactly

$$(3\Delta - 2\alpha_i - 3\beta_i) + 2\alpha_i + 3\beta_i = 3\Delta$$

unlabeled vertices. The k common roots are retained when labels are forgotten, so every constituent has exactly $k(3\Delta + 1)$ vertices.

Let (G, x) be a finite weighted graph and fix a map $\phi : [k] \rightarrow V(G)$. If ϕ does not preserve every adjacency and non-adjacency of H , all the rooted gadget evaluations vanish. Otherwise define

$$U_i(\phi) := \{u \in V(G) : \phi \text{ remains an induced map after replacing } \phi(i) \text{ by } u\}$$

and let μ_i be the total weight of $U_i(\phi)$. When $\mu_i > 0$, normalize the restricted weights on $U_i(\phi)$ and write

$$X_i := t(K_2; G[U_i(\phi)]), \quad Y_i := t(K_3; G[U_i(\phi)]).$$

The definitions give

$$t(V_i; G, x, \phi) = \mu_i, \quad t(E_i; G, x, \phi) = \mu_i^2 X_i, \quad t(T_i; G, x, \phi) = \mu_i^3 Y_i.$$

Consequently, with the value interpreted as zero if some $\mu_i = 0$,

$$t(\psi_H(\tau_\Delta(q)); G, x, \phi) = q(X_1, \dots, X_k, Y_1, \dots, Y_k) \prod_{i=1}^k \mu_i^{3\Delta}.$$

Each (X_i, Y_i) belongs to $\mathcal{R}_{\text{et}}$, proving (i) after averaging over ϕ .

For (ii), replace vertex i of H by a clique of order n_i , and put all or no cross-edges between two cliques according as the corresponding pair is an edge or a non-edge of H . We first check that every induced map ϕ of the labeled copy of H into this blowup is injective. A collision fiber of ϕ is an independent homogeneous set of H : it is independent because the host has no loops, and two labels with the same image have identical adjacency to every label outside the fiber. Thus a proper collision fiber of size at least two contradicts stringency. If all labels collide, then H is edgeless, which contradicts rigidity because $k \geq 6$. Hence every collision fiber is a singleton.

Now the labels mapped into any one blowup clique form a homogeneous set of H . They cannot all lie in one clique: injectivity would then make $H = K_k$, again contradicting rigidity. Hence every nonempty clique-preimage is proper, so stringency makes it a singleton. The k labels therefore occupy the k cliques bijectively, and the resulting permutation is an automorphism of H . Rigidity forces it to be the identity; consequently label i is mapped into the i th clique and $U_i(\phi)$ is exactly that clique. This is the argument of Hatami and Norine [16, Claim 5.8]. Thus every nonzero rooted contribution is the displayed negative value multiplied by a positive factor, and at least one such contribution occurs. The unrooted evaluation is therefore negative. Finally, H has an edge, so there is no induced map of H into the one-vertex edgeless graph; this proves (iii). $\square$

We now derive the required ordinary score function. Fix a standard computably enumerable nonrecursive set $\mathfrak{U} \subseteq \mathbb{N}$, together with an index for its enumeration. The Davis–Putnam–Robinson–Matiyasevich theorem [9, 27, 28] and a bounded-variable universal Diophantine equation of Jones [18], applied to this fixed index, effectively give a fixed integer $k \geq 6$ and a fixed polynomial

$$P(A, Z_1, \dots, Z_k) \in \mathbb{Z}[A, Z_1, \dots, Z_k]$$

such that

$$a \in \mathfrak{U} \iff P(a, n_1, \dots, n_k) = 0 \text{ for some } n_1, \dots, n_k \in \mathbb{N}_{>0}.$$

Here the usual nonnegative variables have been replaced by $n_i - 1$, and dummy variables may be added, so positivity and $k \geq 6$ cause no loss. Put $R_a(Z) := 2P(a, Z)^2 - 1$, and fix the specialization-independent bound

$$D := \max \left\{ 1, \max_{i \in [k]} \deg_{Z_i} (2P(A, Z)^2 - 1) \right\}.$$

Clearing denominators gives the integer polynomial

$$p_a(x_1, \dots, x_k) := \left(\prod_{i=1}^k (1 - x_i)^D \right) R_a \left(\frac{1}{1 - x_1}, \dots, \frac{1}{1 - x_k} \right).$$

At the clique grid,

$$p_a \left(1 - \frac{1}{n_1}, \dots, 1 - \frac{1}{n_k} \right) = \left(\prod_{i=1}^k n_i^{-D} \right) (2P(a, n_1, \dots, n_k)^2 - 1),$$

so this value is negative exactly when the Diophantine equation is satisfied.

The coefficient of $\prod_i (1 - x_i)^D$ in the displayed clearing-denominators expansion is $2P(a, 0)^2 - 1$, an odd nonzero integer. Since $D \geq 1$, the polynomial p_a is nonconstant. Let $\|p_a\|_1$ be the sum of the absolute values of its coefficients and set

$$M_a := 100 \deg(p_a) \|p_a\|_1.$$

The Hatami–Norine penalty polynomial is

$$q_a(x_1, \dots, x_k, y_1, \dots, y_k) := p_a(x_1, \dots, x_k) \prod_{i=1}^k (1 - x_i)^6 + M_a \sum_{i=1}^k (y_i - b(x_i)).$$

Their penalty lemma [16, Lemma 5.4] gives, at points where every $x_i < 1$,

$$\begin{aligned} a \notin \mathfrak{U} &\implies q_a \geq 0 \text{ on } \mathcal{R}_{\text{et}}^k, \\ a \in \mathfrak{U} &\implies q_a \text{ is negative at a clique-grid point.} \end{aligned}$$

If some $x_i = 1$, the first term in q_a vanishes, while $y_i \geq L(x_i) \geq b(x_i)$ for every i , so no new negative point occurs on the boundary. Thus the displayed implication holds on all of $\mathcal{R}_{\text{et}}^k$. All these polynomials have the common effective degree bound

$$\Delta := \max\{2, kD + 6k\},$$

independent of a .

Theorem 6.3. *There exist an effectively computable integer $d_0 \geq 2$, a computably enumerable nonrecursive set $\mathfrak{U} \subseteq \mathbb{N}$, and a computable sequence*

$$g_a = \sum_{F \in \mathcal{U}_{d_0}} \gamma_{a,F} I_{d_0}(F), \quad 0 < \gamma_{a,F} < 1,$$

with rational coefficients, such that

$$\text{Val}_{d_0}(g_a) \geq \frac{1}{2}, \quad a \notin \mathfrak{U} \iff \text{Val}_{d_0}(g_a) = \frac{1}{2}, \quad a \in \mathfrak{U} \iff \text{Val}_{d_0}(g_a) > \frac{1}{2}. \quad (6.1)$$

An even common denominator N_a of the coefficients is computable from a .

Proof. Apply Lemma 6.2 with the fixed parameters k, Δ , put $d_0 := k(3\Delta + 1)$, and write

$$h_a := \mathbf{L}(q_a) = \sum_J c_{a,J} J.$$

Every J in this finite integer quantum graph has exactly d_0 vertices. For $F \in \mathcal{U}_{d_0}$, fix representatives of F and of every J on $[d_0]$, and define

$$b_{a,F} := \frac{1}{d_0!} \sum_{\sigma \in S_{d_0}} \sum_J c_{a,J} \mathbf{1}_{\{E(J) \subseteq E(\sigma F)\}}.$$

Exchangeability of the d_0 sampled positions gives the exact induced-basis identity

$$t(h_a; G, x) = t\left(\sum_{F \in \mathcal{U}_{d_0}} b_{a,F} I_{d_0}(F); G, x\right)$$

for every finite weighted graph. In particular, $b_{a,F} \in (d_0!)^{-1}\mathbb{Z}$, and the entire coefficient vector is effectively computable from a .

Put

$$C_a := 1 + \left\lceil \max_{F \in \mathcal{U}_{d_0}} |b_{a,F}| \right\rceil$$

and define

$$g_a := \frac{1}{2} \mathbf{1}_{d_0} - \frac{1}{2C_a} \sum_{F \in \mathcal{U}_{d_0}} b_{a,F} I_{d_0}(F).$$

Thus $\gamma_{a,F} = 1/2 - b_{a,F}/(2C_a)$ lies strictly between 0 and 1, and $N_a := 2C_a d_0!$ is a computable even common denominator. Moreover,

$$\text{Val}_{d_0}(g_a) = \frac{1}{2} - \frac{1}{2C_a} \inf_{G,x} t(h_a; G, x).$$

If $a \notin \mathfrak{U}$, the penalty property and Lemma 6.2(i),(iii) show that this infimum is zero. If $a \in \mathfrak{U}$, part (ii) gives a finite graph on which h_a is negative. This proves all three assertions in (6.1). $\square$

The constant score table

$$g^\circ := \frac{1}{2} \mathbf{1}_{d_0}$$

has value exactly $1/2$ and may be represented with every even common denominator.

6.2 The ordinary transcendence input

The companion manuscript of Li, Liu, and Liu [21], version dated August 4, 2026, proves the following transcendence result. We use its Theorem 1.2 and Lemmas 3.1–3.3: for the normalized 2×4 input in that proof, Lemma 3.1 gives a finite obstruction basis, Lemma 3.2 computes its class Lagrangian, and Lemma 3.3 gives a finite pair-covering family. This is the only theorem used here without proof.

Theorem 6.4 (Ordinary transcendence construction). *There is an effectively computable integer r_{tr} such that, for every $r \geq r_{\text{tr}}$, one can effectively construct a finitely based monotone class*

$$\mathcal{D}_r^{\text{tr}} = \text{Forb}(\mathcal{A}_r^{\text{tr}})$$

of r -graphs. If $\mathcal{P}_r^{\text{tr}}$ is the finite pair-covering family obtained from $\mathcal{A}_r^{\text{tr}}$ as in Theorem 2.4, then

$$\beta_r^{\text{tr}} := r! \Lambda(\mathcal{D}_r^{\text{tr}}) = \pi(\mathcal{P}_r^{\text{tr}})$$

is transcendental. The edge lists of both finite families are part of the effective output; the construction does not evaluate β_r^{tr} .

7 A rational-power ordinary construction

Throughout this section, the sample order $d \geq 2$ and the desired power $h \geq 1$ are fixed. The target ordinary uniformity is denoted by r and is taken sufficiently large in terms of (d, h) . Put $c_d := 2d!$ and $\tau := 1/2$. Our goal is the following construction theorem.

For $m \geq 1$, write

$$\Delta_m := \left\{ x \in \mathbb{R}^m : x_i \geq 0, \sum_{i=1}^m x_i = 1 \right\}, \quad e_j(x) := \sum_{I \in \binom{[m]}{j}} \prod_{i \in I} x_i.$$

Thus Δ_m is the probability simplex and e_j is the j th elementary symmetric polynomial. We use the same notation for vectors whose coordinates are indexed by a finite set.

Theorem 7.1. *There is a single partial algorithm, uniform in all the displayed finite data, with the following property. For every $d \geq 2$ and $h \geq 1$, there is an integer $r_{\text{pow}}(d, h)$ such that, whenever $r \geq r_{\text{pow}}(d, h)$ and*

$$f = \sum_{F \in \mathcal{U}_d} a_F I_d(F), \quad 0 \leq a_F \leq 1,$$

is a rational d -vertex score table, the algorithm halts. It computes

$$\rho_{r,h} = \theta_{r,h}^h \in \mathbb{Q} \cap (0, 1), \quad \theta_{r,h} \in \mathbb{Q}_{>0},$$

and outputs a finite obstruction family $\mathcal{A}_f$ of r -graphs, the finitely based monotone class $\mathcal{D}_f := \text{Forb}(\mathcal{A}_f)$, and its finite pair-covering extension family

$$\mathcal{F}_f := \bigcup_{A \in \mathcal{A}_f} \text{Ext}(A).$$

For every such f , the output families satisfy

$$\pi(\mathcal{F}_f) = r! \Lambda(\mathcal{D}_f).$$

If $\text{Val}_d(f) \geq 1/2$, then

$$r! \Lambda(\mathcal{D}_f) \geq \rho_{r,h}, \quad r! \Lambda(\mathcal{D}_f) = \rho_{r,h} \iff \text{Val}_d(f) = \frac{1}{2}.$$

Under the same hypothesis, strict inequality holds exactly when $\text{Val}_d(f) > 1/2$. The rational pair $(\theta_{r,h}, \rho_{r,h})$ depends only on (r, h) and is independent of the coefficient vector and its common denominator.

Outline of the construction. The construction uses weights at two levels. After choosing the alphabet and the parts, replace each root part B by the labeled vertices B_a . A weighting has *within-block* coordinates

$$w^B = (w_a^B)_{a \in \Omega_N}$$

inside each root block and *part* coordinates

$$x_B := \sum_{a \in \Omega_N} w_a^B, \quad x_\star := z,$$

where z is the total weight of the data vertices. We first optimize the weights inside each block while fixing the total weight of every part, and then optimize over the vector of part weights.

The proof has six steps.

- (1) Translation relations give the rational score coefficients and make each state or coefficient edge type unique. A regular spectral relation gives a positive correction whose nonconstant operator norm is small.
- (2) We choose D_{del} arithmetically and delete a union of D_{del} Baranyai factors from the complete r -graph on the parts. The value after deletion is the rational number $\rho_{r,h}$.
- (3) Balancing relations give negative curvature inside every root block. Two additional support systems make the gradient in the part weights vanish at the uniform vector while preserving this curvature.
- (4) Rooted state and coefficient edges specify a finite graph on the data vertices and record the score of each induced d -vertex type.
- (5) A bounded witness tree removes the auxiliary root labels and gives a finite obstruction family of ordinary r -graphs.
- (6) We prove that the weights are uniform inside each root block, complete the data-pair relation, and reduce the class Lagrangian to a scalar optimization. At the reference value, the uniform vector of part weights is the unique maximizer. The pair-covering transfer then gives the required finite forbidden family.

Order of choices. The table records the main groups of objects and which earlier objects they use.

Step	Objects	Chosen after	Purpose
Input score	$d, f = \sum_F a_F I_d(F)$, and a common denominator N	the finite input	input objective
Finite alphabet	$\Omega_N, q_N, \eta, \mathbf{X}_{q_N}, S_F, T_N, Q_N, \mathbf{U}_N$	d, N , and $(a_F)_F$	score coefficients and completion of the data relation
Reference value	$M, P, b_{r,h}, u, \theta_{r,h}, \rho_{r,h}, D_{\text{del}}$	r, h only	reference value and deletion degree
Design on the parts	$K, R, \mathcal{H}, \mathcal{C}_{\text{sub}}, \mathcal{C}_{\text{add}}, \mathcal{E}_{\text{pow}}$	d, h, r only	balance and gradient cancellation
Final families	$\mathcal{L}_N, \mathcal{D}_f, \mathcal{A}_f, \mathcal{F}_f$	all finite input data	removal of root labels and finite ordinary families

In particular, the reference pair $(\theta_{r,h}, \rho_{r,h})$ is independent of the coefficient relations and depends only on (r, h) . The score table affects the output family, but not its equality reference value.

7.1 Translation relations and a spectral relation

Let N be a common denominator of the coefficients of an input f . Choose effectively a finite product of finite fields Ω_N such that, with $q_N := |\Omega_N|$,

$$c_d N \mid q_N, \quad (7.1)$$

every field factor has order at least 4, and q_N is larger than a constant specified below. Enlarging field exponents or adjoining extra field factors preserves (7.1), so this is always possible. Choose $\eta \in \Omega_N$ such that $0, 1, \eta$ are pairwise distinct in every field factor. Their pairwise differences are units.

For $S \subseteq \Omega_N$, define the *three-block translation relation*

$$\mathbf{D}(S) := \{(b, b + \xi, b + \eta\xi) : b \in \Omega_N, \xi \in S\}. \quad (7.2)$$

For nonnegative vectors $u^{(1)}, u^{(2)}, u^{(3)}$ on Ω_N , let $T_S(u^{(1)}, u^{(2)}, u^{(3)})$ be the corresponding *trilinear form*.

The translation relation has the following counting and norm properties.

Lemma 7.2. *For every $S \subseteq \Omega_N$:*

- (i) $|\mathbf{D}(S)| = q_N |S|$;
- (ii) *projection to any two coordinates is injective*;

(iii) if the three total masses are U_1, U_2, U_3 , then

$$0 \leq T_S \leq U_1 U_2 U_3, \quad \left\| \frac{\partial^2 T_S}{\partial u^{(i)} \partial u^{(j)}} \right\|_{\text{op}} \leq U_k$$

for distinct i, j, k ;

(iv) at uniform internal weights,

$$T_S = \frac{|S|}{q_N^2} U_1 U_2 U_3. \quad (7.3)$$

Every coordinate class is regular in the relation.

Proof. Two coordinates determine ξ because the corresponding difference of $0, 1, \eta$ is a unit, and then determine b . This proves the first two statements and regularity. In a mixed Hessian every matrix entry is zero or the weight of the uniquely determined complementary coordinate. Every row and column sum is at most the complementary total mass, so $\|A\|_{\text{op}} \leq \sqrt{\|A\|_1 \|A\|_\infty}$ gives the operator bound. The uniform evaluation follows from the cardinality. $\square$

The positive correction uses a relation with a much smaller nontrivial operator norm than a union of matchings.

Lemma 7.3. *There is an absolute integer $D_{\text{sp}} \geq 3$ such that, for every sufficiently large q , there is a simple D_{sp} -regular bipartite graph $\mathbf{X}_q$ with two vertex classes of order q and*

$$\sigma_2(\mathbf{X}_q) \leq \frac{D_{\text{sp}}}{16}, \quad (7.4)$$

where σ_2 is the largest singular value on the orthogonal complements of the two constant vectors. Moreover, if $q_1 < q_2 < \dots$ is any computable unbounded sequence, then one can effectively find an index j and such a graph $\mathbf{X}_{q_j}$.

Proof. Take, for example, $D_{\text{sp}} = 4096$. The equal-side specialization of the simple bipartite biregular model in Brito, Dumitriu, and Harris [7, Theorem 3.2(i)] states directly that a random simple D_{sp} -regular bipartite graph has every nontrivial singular value at most $2\sqrt{D_{\text{sp}} - 1} + o(1)$ with probability tending to one; this is the bipartite analogue of the results of Friedman [12] and Bordenave [6]. Since $2\sqrt{D_{\text{sp}} - 1} < D_{\text{sp}}/16$, the required graph exists for every sufficiently large q .

For effectivity, at step j enumerate the finitely many simple bipartite graphs with two classes of order q_j , retain the D_{sp} -regular ones, and compare their singular values with $D_{\text{sp}}/16$ using the exact real-algebraic algorithms of Basu, Pollack, and Roy [4]. Dovetail these finite searches. Eventual existence guarantees termination; no numerical spectral threshold is needed by the construction. $\square$

Starting from the alphabet already chosen, consider

$$\Omega_N^{(j)} := \Omega_N \times \mathbb{F}_5^j \quad (j = 0, 1, 2, \dots)$$

and apply Lemma 7.3 to the resulting computable sequence of alphabet orders. Choose the first j for which the required spectral relation is found. From now on, replace Ω_N by $\Omega_N^{(j)}$, replace η by

$$(\eta, 2, \dots, 2) \in \Omega_N \times \mathbb{F}_5^j,$$

and reset $q_N := |\Omega_N|$. Thus (7.1) is preserved and $0, 1, \eta$ are still componentwise distinct. We also redefine $D(S)$ and T_S using this final alphabet; the proof of Lemma 7.2 applies verbatim.

Only now choose the coefficient and unit subsets. For $a_F = m_F/N$, take $S_F \subseteq \Omega_N$ with $|S_F| = a_F q_N$, put

$$Q_N := c_d q_N, \quad (7.5)$$

choose $T_N \subseteq \Omega_N$ with $|T_N| = q_N/c_d$, and define

$$U_N := D(T_N). \quad (7.6)$$

At uniform internal weights,

$$\frac{|\mathbf{U}_N|}{q_N^3} = \frac{1}{Q_N}. \quad (7.7)$$

This order of choices ensures that all cardinalities refer to the final alphabet.

Fix the relation $\mathbf{X}_{q_N}$ found above. If nonnegative vectors u, v have totals U, V , then at uniform weights

$$\sum_{xy \in E(\mathbf{X}_{q_N})} u_x v_y = \frac{D_{\text{sp}}}{q_N} UV. \quad (7.8)$$

On the product of the two zero-sum tangent spaces, the mixed Hessian has operator norm at most $D_{\text{sp}}/16$.

7.2 The rational power base

Put

$$C_h := 20(h+1), \quad M := r^{C_h}, \quad P := M-1, \quad (7.9)$$

so $r \mid M$. Let

$$A_{M,r} := \frac{(M)_r}{M^r} = r! \binom{M}{r} M^{-r}. \quad (7.10)$$

Set

$$b_{r,h} := \left\lfloor \frac{r-1}{h} \right\rfloor, \quad L_{r,h} := (r-1)!. \quad (7.11)$$

In particular, $(r-1)! \mid L_{r,h}^h$.

We first choose the arithmetic parameter that fixes the rational reference value.

Lemma 7.4. *For all sufficiently large r , with an effective lower bound depending on (d, h) , there is a positive multiple u of $L_{r,h}$ satisfying*

$$M^{b_{r,h}}(A_{M,r} - 2M^{-5})^{1/h} \leq u \leq M^{b_{r,h}}(A_{M,r} - M^{-5})^{1/h}. \quad (7.12)$$

Define

$$\theta_{r,h} := \frac{u}{M^{b_{r,h}}}, \quad \rho_{r,h} := \theta_{r,h}^h. \quad (7.13)$$

The constant $\theta_{r,h}$ sets the rational reference value and is unrelated to the alphabet size q_N from (7.1). Below, $z = x_\star$ denotes total data mass, whereas $\rho_{r,h}$ denotes the fixed rational reference value in (7.13). One has $0 < \rho_{r,h} < 1$, and

$$D_{\text{del}} := \frac{M^{r-1}}{(r-1)!} (A_{M,r} - \rho_{r,h}) \quad (7.14)$$

is a positive integer satisfying

$$\frac{M^{r-6}}{(r-1)!} \leq D_{\text{del}} \leq \frac{2M^{r-6}}{(r-1)!}. \quad (7.15)$$

All quantities are effectively computable from (r, h) .

Proof. The product estimate

$$A_{M,r} = \prod_{i=0}^{r-1} \left(1 - \frac{i}{M}\right) \geq 1 - \frac{\binom{r}{2}}{M}$$

gives $A_{M,r} > 1/2$ once $M > r(r-1)$. On $(0, 1)$ the derivative of $x^{1/h}$ is at least $1/h$. The two arguments of the roots in (7.12) lie in $(0, 1)$ for all sufficiently large r under the same stated conditions, so that interval has length at least $M^{b_{r,h}-5}/h$. If $r \geq 12h+2$, then

$$b_{r,h} - 5 \geq \frac{r}{2h}, \quad \frac{M^{b_{r,h}-5}}{h} \geq \frac{r^{10r}}{h}.$$

After also requiring $r^{9r+1} > h$, this is greater than $r^{r-1} \geq (r-1)! = L_{r,h}$. Thus the interval contains a positive multiple of $L_{r,h}$; it is found by comparing h th powers of integers with the rational endpoints. These inequalities prove the assertion for every later r , rather than only for a searched value.

Since $A_{M,r} = (M-1)_{r-1}/M^{r-1}$,

$$D_{\text{del}} = \frac{(M-1)_{r-1} - u^h M^{r-1-b_{r,h}h}}{(r-1)!}.$$

The exponent $r-1-b_{r,h}h$ is nonnegative and $(r-1)! \mid u^h$, proving integrality. The bounds follow immediately from (7.12). $\square$

Because $r \mid M$, Baranyai's factorization theorem [3] gives a decomposition of the complete r -graph on the M part coordinates,

$$\mathfrak{B}_{M,r} = \{\mathcal{M}_1, \dots, \mathcal{M}_{T_{M,r}}\}, \quad T_{M,r} = \binom{M-1}{r-1}, \quad (7.16)$$

into perfect matchings. Fix one such decomposition effectively, for example by exhaustive search through the finite set of edge partitions.

Put $K_0 := 64(c_d D_{\text{sp}} + 1)$ and $K := 2D_{\text{sp}}K_0$, and fix the explicit reservoir constant $C_d^{\text{res}} := K + c_d D_{\text{sp}} + c_d K/2 + 4d! + K_0 + 3$.

The following estimate shows that enough matching factors remain after all bounded exclusions.

Lemma 7.5. *Let Z be a set of at most $5r$ part coordinates, and let $\mathfrak{B}_Z^{\text{bad}}$ be the factors having an edge that meets Z in at least two vertices. Then*

$$|\mathfrak{B}_Z^{\text{bad}}| \leq \binom{|Z|}{2} \binom{M-2}{r-2}.$$

Let $\mathfrak{E}$ be any further family of excluded factors and $\mathfrak{P}$ any family of pairwise distinct prescribed factors satisfying

$$|\mathfrak{E}| \leq C_d^{\text{res}}(M + r^2), \quad |\mathfrak{P}| \leq C_d^{\text{res}} r,$$

and assume

$$\mathfrak{P} \cap (\mathfrak{B}_Z^{\text{bad}} \cup \mathfrak{E}) = \emptyset, \quad |\mathfrak{P}| \leq D_{\text{del}}. \quad (7.16a)$$

For all sufficiently large r , there is a family of exactly D_{del} factors outside $\mathfrak{B}_Z^{\text{bad}} \cup \mathfrak{E}$ that contains $\mathfrak{P}$.

Proof. Every bad factor contains an r -edge containing a selected pair from Z ; each such edge lies in exactly one Baranyai factor. This gives the stated bound and hence

$$\frac{|\mathfrak{B}_Z^{\text{bad}}|}{T_{M,r}} \leq \binom{|Z|}{2} \frac{r-1}{M-1}.$$

For all sufficiently large r satisfying an explicit bound, $M \geq 4r$, so

$$T_{M,r} = \binom{M-1}{r-1} \geq \left(\frac{M}{2r}\right)^{r-1}.$$

Thus

$$\frac{|\mathfrak{E}|}{T_{M,r}} \leq C_d^{\text{res}}(M + r^2) \left(\frac{2r}{M}\right)^{r-1}.$$

On the other hand, by (7.15) and $A_{M,r} > 1/2$,

$$\frac{D_{\text{del}}}{T_{M,r}} = \frac{A_{M,r} - \rho_{r,h}}{A_{M,r}} \leq 4M^{-5}.$$

For later reference, an exact sufficient availability condition is

$$T_{M,r} - \binom{5r}{2} \binom{M-2}{r-2} - C_d^{\text{res}}(M + r^2) \geq D_{\text{del}}. \quad (7.16b)$$

Indeed, the left side is a lower bound for the number of factors outside $\mathfrak{B}_Z^{\text{bad}} \cup \mathfrak{E}$. After division by $T_{M,r}$, the three terms that have to be subtracted or retained are bounded by

$$\binom{5r}{2} \frac{r-1}{M-1}, \quad C_d^{\text{res}}(M + r^2) \left(\frac{2r}{M}\right)^{r-1}, \quad 4M^{-5},$$

respectively. Their sum is less than one for all sufficiently large r , as verified by exact integer and rational comparisons. Thus (7.16b) holds for all such r . At least D_{del} factors are then available; retain $\mathfrak{P}$ and choose any further $D_{\text{del}} - |\mathfrak{P}|$ available factors. Scanning the fixed finite decomposition makes this choice effective. $\square$

A union $\mathcal{E}_{\text{pow}}$ of D_{del} factors of $\mathfrak{B}_{M,r}$ is D_{del} -regular. At the uniform part-weight vector

$$p := (M^{-1}, \dots, M^{-1}), \quad (7.17)$$

we have

$$r! \left(e_r(p) - \sum_{E \in \mathcal{E}_{\text{pow}}} \prod_{i \in E} p_i \right) = A_{M,r} - \frac{(r-1)! D_{\text{del}}}{M^{r-1}} = \rho_{r,h}. \quad (7.18)$$

The matching reservoir will allow us to choose $\mathcal{E}_{\text{pow}}$ after all other supports on the parts have been fixed.

7.3 Balancing relations and correction supports

Put

$$R := \frac{rK}{2}. \quad (7.19)$$

Thus K is even and $R/D_{\text{sp}} = rK_0$ is an integer. Write $\mathbb{Z}_P := \mathbb{Z}/P\mathbb{Z}$, with addition taken modulo P .

The balancing relations require a regular support system whose oriented profiles are pairwise distinct.

Lemma 7.6. *For all sufficiently large r , there is an effectively constructible family*

$$\mathcal{H} = \{(E, \partial E) : E \in \binom{\mathbb{Z}_P}{r}, \partial E \in \binom{\mathbb{Z}_P}{2}\}$$

such that:

- (i) *the supports E are pairwise distinct;*
- (ii) *every root part belongs to exactly R supports;*
- (iii) *every root part belongs to exactly K distinguished pairs;*
- (iv) *the oriented profiles $(B, E \setminus \partial E)$, with $B \in \partial E$, are pairwise distinct.*

Proof. Write $K = 2k$. For $v \in [k]$, put

$$A_v := 4rv, \quad D_v := 4r(k+v+1), \quad L_v := A_v + D_v$$

and

$$S_v := \{0, L_v\} \cup \{A_v + j : 1 \leq j \leq r-2\} \subseteq \mathbb{Z}_P.$$

For all sufficiently large r , we may assume that $r \geq 3$ and

$$P > 2L_k = 8r(3k+1).$$

For every $x \in \mathbb{Z}_P$, take $E_{x,v} = x + S_v$ and $\partial E_{x,v} = \{x, x + L_v\}$. With the elements of S_v represented in $\{0, \dots, P-1\}$, we have

$$0 < A_v + 1 < \dots < A_v + r - 2 < L_v < P.$$

The cyclic gaps of S_v have lengths $A_v + 1$, 1 for the internal gaps, $D_v - r + 2$, and $P - L_v$. The last is uniquely largest: indeed, $P - L_v \geq P - L_k > L_k$, whereas each of the other gap lengths is less than $L_v \leq L_k$. Consequently, if $E_{x,v} = E_{y,w}$, the unique largest gap identifies both its successor $x = y$ and its predecessor $x + L_v = y + L_w$. Since

$$L_v = 4r(k + 2v + 1)$$

is strictly increasing in v , this gives $v = w$. Thus the supports are pairwise distinct.

It remains to verify the oriented assertion. Subtracting the distinguished vertex B from the residual support gives

$$(E_{x,v} \setminus \partial E_{x,v}) - B = \begin{cases} I_v^+ := \{A_v + 1, \dots, A_v + r - 2\}, & B = x, \\ I_v^- := \{1 - D_v, \dots, r - 2 - D_v\} \pmod{P}, & B = x + L_v. \end{cases}$$

The representatives of I_v^+ in $\{0, \dots, P-1\}$ lie strictly between 0 and L_k . Since $D_v > r-2$ and $D_v < L_k$, those of I_v^- are

$$\{P+1-D_v, \dots, P+r-2-D_v\},$$

and lie strictly between L_k and P . Hence no I_v^+ equals an I_w^- . Within either family the intervals are pairwise disjoint, because

$$A_{v+1} - A_v = D_{v+1} - D_v = 4r > r-3.$$

Thus all $2k$ residual-offset sets are distinct. Equality of two oriented profiles first gives the same distinguished vertex B and then the same residual-offset set, hence the same choice of sign and the same v ; finally $x = B$ in the first case and $x = B - L_v$ in the second. This proves (iv).

For each fixed v , every part occurs in exactly r of the supports $E_{x,v}$ and occurs once as each of the two distinguished endpoints. Summing over $v \in [k]$ gives support degree $rk = rK/2 = R$ and distinguished degree $2k = K$, proving (ii) and (iii). $\square$

The root parts will contain all special blocks. Reserve disjoint root parts as follows:

- for each $j \in [4]$, two state blocks A_j, B_j and a filler set $F_{S,j}$ of $r-4$ root parts;
- three translation blocks U_1, U_2, U_3 and a coefficient filler set F_P of $r-d-3$ root parts.

The number of special root parts is

$$s_{\text{special}} = 4(r-2) + (r-d) = 5r-d-8. \quad (7.20)$$

All other root parts are padding. The data coordinate is denoted by $\star$ and is identified with the remaining point of the M -set. Put

$$h_{\star} := 8d! + d, \quad h_B := \begin{cases} d!, & B \text{ is a state block or state filler,} \\ 1, & B \text{ is a translation block or coefficient filler,} \\ 0, & B \text{ is padding,} \end{cases} \quad (7.21)$$

and

$$H_{\text{special}} := \sum_B h_B = 4(r-2)d! + (r-d). \quad (7.22)$$

Finally set

$$b_0 := c_d D_{\text{sp}}, \quad L := c_d R + c_d D_{\text{sp}}, \quad d_+ := \frac{L}{c_d D_{\text{sp}}} = rK_0 + 1. \quad (7.23)$$

The constants above are chosen to satisfy the analytic estimates below. At the uniform part-weight vector p , after multiplication by $Q_N M^{r-1}$, the balancing, state, and coefficient terms introduced below have derivatives

$$h_{\star} \quad \text{at } \star, \quad -c_d R + h_B \quad \text{at a root part } B;$$

see Lemma 7.15. Since $L = c_d R + b_0$, the degree prescription in (7.24) is given by these derivatives plus the same value L in every coordinate:

$$b_0 + c_d R + h_{\star} = h_{\star} + L, \quad b_0 + h_B = (-c_d R + h_B) + L.$$

A negative unit correction contributes minus its support degree, whereas a d_+ -regular positive correction contributes $c_d D_{\text{sp}} d_+ = L$ in every coordinate. Thus the target degree identities give the coordinatewise cancellation

$$(\text{derivative before corrections})_i - d_{\mathcal{C}_{\text{sub}}}(i) + c_d D_{\text{sp}} d_+ = 0.$$

The bounded designation loads serve a different purpose: they keep the within-block Hessian perturbations below the curvature given by the balancing relations. Finally, the regular union $\mathcal{E}_{\text{pow}}$ preserves the uniform gradient with respect to the part weights and gives the power value in (7.18).

The next lemma constructs the two correction support systems and the power deletion simultaneously.

Lemma 7.7. *For every sufficiently large r , one can effectively choose simple r -graphs $\mathcal{C}_{\text{sub}}, \mathcal{C}_{\text{add}}, \mathcal{E}_{\text{pow}}$ on the M part coordinates, together with designated root coordinates on their edges, so that:*

(i) $\mathcal{C}_{\text{sub}}$ is disjoint from the balancing supports and from $\mathcal{E}_{\text{pow}}$, and

$$d_{\mathcal{C}_{\text{sub}}}(\star) = b_0 + c_d R + h_\star, \quad d_{\mathcal{C}_{\text{sub}}}(B) = b_0 + h_B; \quad (7.24)$$

- (ii) every edge of $\mathcal{C}_{\text{sub}}$ has three designated padding coordinates, and every padding coordinate is designated on at most b_0 edges;
- (iii) $\mathcal{C}_{\text{add}}$ is the union of d_+ perfect-matching factors, is d_+ -regular, is disjoint from $\mathcal{C}_{\text{sub}}$ and the balancing supports, and is contained in $\mathcal{E}_{\text{pow}}$;
- (iv) every edge of $\mathcal{C}_{\text{add}}$ has two designated root coordinates, and every root coordinate is designated on at most

$$T_+ := 2K_0 + 5 \quad (7.25)$$

edges;

- (v) $\mathcal{E}_{\text{pow}}$ is a union of D_{del} perfect-matching factors, contains $\mathcal{C}_{\text{add}}$, is disjoint from $\mathcal{C}_{\text{sub}}$ and the balancing supports, and is D_{del} -regular.

Proof. Step 1: reserve clean Baranyai factors. Let Z consist of $\star$ and the special root parts. By Lemma 7.5, exclude every factor containing a balancing support; there are at most $|\mathcal{H}| = PK/2$ such factors. Under the stated lower bounds on r , $D_{\text{del}} \geq C_d^{\text{res}} r \geq b_0 + d_+$. Choose $b_0 + d_+$ factors from the resulting reservoir. They are pairwise distinct, every edge meets Z in at most one vertex, and no edge is a balancing support. Reserve the first b_0 as baseline factors and the remaining d_+ as the factors of $\mathcal{C}_{\text{add}}$.

Step 2: realize the negative-correction degrees. Start $\mathcal{C}_{\text{sub}}$ with the union of the baseline factors. Every part coordinate initially has degree b_0 . We must add $c_d R + h_\star$ incidences at $\star$ and h_B incidences at every special root part. Put

$$S_c := c_d R + h_\star + H_{\text{special}}. \quad (7.26)$$

The divisibility needed for the switching construction is immediate: using $c_d R = r(c_d K/2)$ and (7.21)–(7.22),

$$S_c = c_d R + h_\star + H_{\text{special}} = r \left(\frac{c_d K}{2} + 4d! + 1 \right),$$

so $r \mid S_c$.

Each baseline factor has exactly M/r edges and at most $|Z|$ of them meet Z . For large r it therefore has more than $(r-1)S_c/r$ edges consisting only of padding coordinates. Remove exactly $(r-1)S_c/r$ such edges from the first baseline factor and let W be their union; then $|W| = (r-1)S_c$. Form a multiset of S_c core incidences, consisting of $c_d R + h_\star$ copies of $\star$ and h_B copies of each special root part. Partition W into labeled $(r-1)$ -sets and attach one cell to each core incidence.

The partition can be chosen so that no resulting edge is a balancing support or an edge of any of the other selected baseline or positive factors. For a fixed labeled core incidence there are at most $R + b_0 + d_+$ forbidden padding sets: at most R come from balancing supports, and each selected factor contributes at most one. In a uniformly random labeled partition, its cell is uniform among the $\binom{|W|}{r-1}$ subsets of size $r-1$. Hence the union bound gives

$$\Pr(\text{some forbidden attachment}) \leq \frac{S_c(R + b_0 + d_+)}{\binom{(r-1)S_c}{r-1}}. \quad (7.26a)$$

Here

$$S_c = r \left(\frac{c_d K}{2} + 4d! + 1 \right), \quad |W| = (r-1)S_c,$$

and

$$\binom{(r-1)S_c}{r-1} \geq S_c^{r-1}.$$

The right side of (7.26a) is therefore less than one for all sufficiently large r . All labeled partitions form a finite, effectively enumerable set, so exhaustive search finds a valid one for each such r .

Every padding coordinate in W loses one baseline incidence and gains one core incidence. All other padding coordinates retain degree b_0 . The special-part and data-part degrees are exactly (7.24). Every

edge has at most one member of Z , hence at least $r - 1$ padding coordinates; designate any three. The designation load is at most the total degree b_0 of a padding coordinate. This proves (i)–(ii).

Step 3: designate the positive corrections with bounded load. It remains to designate two root coordinates on every edge of the d_+ positive factors with bounded load. Form the bipartite incidence graph whose left side is the set of their support edges and whose right side is the set of root part coordinates; a left vertex is adjacent to the root coordinates on its edge. Every left vertex has degree at least $r - 1$ and every right vertex has degree d_+ . Put $T := \lceil 2d_+/(r - 1) \rceil$. We use the integral b -matching formulation, with demand 2 at every left vertex, capacity T at every right vertex, and capacity 1 on every incidence edge. For a set $\mathcal{X}$ of left vertices, let $d_{\mathcal{X}}(y)$ be the number of its neighbors at the right vertex y . The max-flow cut criterion is

$$\sum_y \min\{T, d_{\mathcal{X}}(y)\} \geq 2|\mathcal{X}| \quad \text{for every } \mathcal{X}. \quad (7.27)$$

Indeed, after $\mathcal{X}$ is fixed, minimizing a cut over the right vertices chooses independently between the T -capacity edge at y and the $d_{\mathcal{X}}(y)$ incidence edges.

Every right vertex has total degree d_+ , and $T \leq d_+$ for all large r . Hence, for $0 \leq a \leq d_+$,

$$\min\{T, a\} \geq \frac{T}{d_+} a.$$

Since every left vertex has degree at least $r - 1$,

$$\sum_y \min\{T, d_{\mathcal{X}}(y)\} \geq \frac{T}{d_+} \sum_y d_{\mathcal{X}}(y) \geq \frac{T(r - 1)}{d_+} |\mathcal{X}| \geq 2|\mathcal{X}|.$$

Thus (7.27) holds. Integral max flow selects two distinct neighbors of every left vertex while using each root coordinate at most T times. For large r , $T \leq 2K_0 + 5 = T_+$. These are the designated blocks of $\mathcal{C}_{\text{add}}$.

Step 4: complete the power deletion. Finally exclude from $\mathfrak{B}_{M,r}$ the b_0 baseline factors, every factor containing a newly created core edge, and every factor containing a balancing support. The number excluded is at most $b_0 + S_c + PK/2 \leq C_d^{\text{res}}(M + r^2)$. The d_+ positive factors meet Z in at most one coordinate, are disjoint from this excluded set, and satisfy $d_+ \leq C_d^{\text{res}} r \leq D_{\text{del}}$. By Lemma 7.5, they can be extended to a set of exactly D_{del} available factors. Their union is $\mathcal{E}_{\text{pow}}$. It contains $\mathcal{C}_{\text{add}}$ and is disjoint from $\mathcal{C}_{\text{sub}}$ and all balancing supports. Distinct factors are edge-disjoint, so all support systems are simple. $\square$

7.4 Rooted edge types and finite obstructions

Before giving the rooted description, we summarize the roles of its edge types.

- The complete squarefree base is the symmetric background objective. Deleting the supports in $\mathcal{E}_{\text{pow}}$ sets the rational reference value.
- A balancing edge repeats one root part. Its e_2 term provides the negative curvature that forces uniform weights inside that root block.
- A family of state edges assigns state 0 or state 1 to a data pair. The four copies ensure that, when the data relation is completed, the gain from completing a missing state dominates every coefficient orbit that may have to be deleted.
- A coefficient orbit attaches a parameter ξ to a data d -set. The translation relation determines ξ uniquely, is regular in every root coordinate, and at uniform root weights gives exactly the coefficient factor prescribed by S_F .
- The deleted $\mathcal{C}_{\text{sub}}$ relations and restored $\mathcal{C}_{\text{add}}$ relations are the negative and positive gradient corrections described above. Their designation-load bounds ensure that they do not destroy the curvature given by the balancing relations.

The filler blocks in the state and coefficient edge types only make every displayed edge r -uniform. The labels and roots are temporary devices in the construction; the output objects are ordinary uncolored r -graphs.

Let

$$\mathcal{L}_N := \bigsqcup_{B \in \mathbb{Z}_P} \{B_a : a \in \Omega_N\}$$

be the finite *root-label set*. A *partial rooting* of an r -graph G is an injective map

$$\mu : \text{dom } \mu \longrightarrow V(G), \quad \text{dom } \mu \subseteq \mathcal{L}_N.$$

The vertices in $\text{im } \mu$ are *rooted vertices*, and the vertices in $V(G) \setminus \text{im } \mu$ are *data vertices*. A rooting is *total* if $\text{dom } \mu = \mathcal{L}_N$. A rooting ν *extends* μ if $\text{dom } \mu \subseteq \text{dom } \nu$ and $\nu|_{\text{dom } \mu} = \mu$.

Every edge type below is interpreted relative to μ : each displayed root label is replaced by its image, every displayed data vertex is chosen from $V(G) \setminus \text{im } \mu$, and all vertices in one displayed edge are distinct. An edge type is *available* when all its displayed root labels lie in $\text{dom } \mu$. An edge is *compatible with* μ if it has one of the available edge types below.

The *root block* corresponding to a root part B is $\{B_a : a \in \Omega_N\}$, whose elements are labeled root vertices. The unrooted vertices form the *data part* $\star$. A total rooting therefore has a *vector of part weights*

$$x = (z, (x_B)_B) \in \Delta_M, \quad (7.28)$$

where z is the total data mass. The edge types below describe ordinary uncolored r -edges; the labels are used only in the finite rooted description.

For distinct data vertices z, w , an index $j \in [4]$, a *state* $\epsilon \in \{0, 1\}$, and $a \in \Omega_N$, take

$$\{z, w, A_{j,a}, B_{j,a+\epsilon}\} \cup \{C_{cc} : C \in F_{S,j}\}, \quad (7.29)$$

where the filler labels vary freely. The *complete state- ϵ family* for the pair $\{z, w\}$ contains all these edges over all four values of j , all a , and all filler labels. For a data d -set X , a parameter $\xi \in \Omega_N$, and $b \in \Omega_N$, take

$$X \cup \{U_{1,b}, U_{2,b+\xi}, U_{3,b+\eta\xi}\} \cup \{C_{cc} : C \in F_P\}. \quad (7.30)$$

The *complete coefficient orbit with parameter* ξ contains all choices of b and all filler labels.

The *squarefree root/data base* is modified as follows.

- (B1) On a balancing support $(E, \{B, B'\}) \in \mathcal{H}$, delete the complete squarefree profile and insert instead every edge using two distinct roots from B and one root from every part of $E \setminus \{B, B'\}$, together with the analogous family using two roots from B' .
- (B2) On a support in $\mathcal{E}_{\text{pow}}$, delete the complete squarefree profile. If the support belongs to $\mathcal{C}_{\text{add}}$, restore precisely the tuples for which the labels on its two designated root blocks form an edge of $\mathbf{X}_{q_N}$; all other labels, and the data vertex when $\star$ is present, are unrestricted.
- (B3) On a support in $\mathcal{C}_{\text{sub}}$, retain the complete squarefree profile except for the tuples whose labels on the three designated padding blocks belong to Ω_N .
- (B4) Every remaining squarefree support is complete.

The next lemma shows that an edge has at most one of these types.

Lemma 7.8. *Relative to a fixed partial rooting, every r -set has at most one type among the allowed base, balancing, state, coefficient, and positive-correction edges. It also determines at most one negative-correction deletion certificate, and these possibilities are pairwise disjoint.*

Proof. Residual base and correction tuples are squarefree at the level of the parts and use at most one data vertex. Rules (B2)–(B4) determine their status uniquely from the support on the parts: $\mathcal{C}_{\text{sub}}$ is disjoint from $\mathcal{E}_{\text{pow}}$, while $\mathcal{C}_{\text{add}} \subseteq \mathcal{E}_{\text{pow}}$ has its fixed restored relation. All these support systems are simple. A balancing edge uses no data vertex and repeats exactly one root part. Its repeated block and singleton profile recover the unique oriented balancing relation by Lemma 7.6(iv).

A state edge has exactly two data vertices and uses the blocks A_j, B_j ; their labels recover (j, ϵ, a) . A coefficient edge has exactly d data vertices and uses U_1, U_2, U_3 ; any two translation labels recover (b, ξ) by Lemma 7.2(ii). If $d \neq 2$, state and coefficient edges already have different data multiplicities. If $d = 2$, the reserved state blocks and translation blocks are disjoint, so their root signatures still separate them. Finally, the designated blocks are fixed on every correction support, and pair-projection injectivity gives a uniquely determined relation tuple. This proves the claim. $\square$

For a partial rooting μ , let E_μ^ϵ be the data-pair relation defined by state- ϵ edges and let P_μ^ξ be the data- d -set relation defined by coefficient edges with parameter ξ . The rooting is *admissible* if every edge has one of the listed edge types and:

$$(S1) \ E_\mu^0 \cap E_\mu^1 = \emptyset;$$

$$(S2) \text{ for every data } d\text{-set } X, \text{ parameter } \xi \in \Omega_N, \text{ and } F \in \mathcal{U}_d,$$

$$\left. \begin{array}{l} X \in P_\mu^\xi, \\ \binom{X}{2} \subseteq E_\mu^0 \cup E_\mu^1, \\ \left[\left(X, E_\mu^1 \cap \binom{X}{2} \right) \right] = F \end{array} \right\} \implies \xi \in S_F. \quad (7.30a)$$

Let $\mathcal{D}_f$ be the class of finite r -graphs admitting an admissible partial rooting.

We now replace the rooted description by a finite ordinary obstruction family.

Lemma 7.9. *The class $\mathcal{D}_f$ is monotone and has an effectively computable finite ordinary obstruction family $\mathcal{A}_f$.*

Proof. The proof has three steps. Every local failure has a bounded witness. Such a witness persists under an extended rooting unless a previously unused root label is mapped into it. Since each extension uses a new label, a finite-depth search tree produces a bounded ordinary obstruction.

Under vertex or edge deletion, restrict the rooting to labels whose images remain. Every surviving compatible edge keeps the same edge type, while all declaration relations can only shrink. Thus (S1) cannot be created, and deletion can only destroy the antecedent of (S2). Hence $\mathcal{D}_f$ is subgraph-closed.

A failure of an edge-type condition, including a forbidden correction tuple determined by the edge, is witnessed by one edge; an (S1) failure is witnessed by two state edges; and an (S2) failure is witnessed by one coefficient edge together with at most one state edge for each of the $\binom{d}{2}$ data pairs. Thus

$$w := r \left(1 + \binom{d}{2} \right)$$

bounds the number of vertices in every witness: one incompatible edge uses at most $r \leq w$ vertices, two state edges use at most $2r \leq w$, and an (S2) witness uses at most $1 + \binom{d}{2}$ edges.

These witnesses have the following persistence property. Let $W = W(G, \mu)$ be the chosen witness, let $H \subseteq G$ contain both W and $\text{im } \mu$, and let ν be a partial rooting of H extending μ . If

$$\nu(\text{dom } \nu \setminus \text{dom } \mu) \cap (V(W) \setminus \text{im } \mu) = \emptyset,$$

then every vertex of W has the same root/data status under μ and ν . By Lemma 7.8, every old edge in W keeps its edge type. Thus a failure of an edge-type condition or state conflict persists. In an (S2) witness, the coefficient edge and all state declarations used to determine its graph type persist. A new declaration on one of those pairs either agrees, leaving the type unchanged, or creates an (S1) conflict. Hence ν cannot be admissible. Consequently every admissible extension must map a previously unused root label into W .

Put $L_{\text{root}} := |\mathcal{L}_N|$. For a nonmember $G \notin \mathcal{D}_f$, build a rooted search tree starting with the empty map. At a node μ , choose a witness W_μ and branch over every one-label injective extension $\mu \cup \{\lambda \mapsto v\}$ with $\lambda \notin \text{dom } \mu$ and $v \in V(W_\mu) \setminus \text{im } \mu$. The depth is at most L_{root} , the branching factor is at most $L_{\text{root}}w$, and the union G_* of all selected witnesses has at most

$$w \sum_{j=0}^{L_{\text{root}}} (L_{\text{root}}w)^j \quad (7.30b)$$

vertices. Along every branch, the image of the current rooting is contained in the union of the witnesses at its ancestor nodes: this holds initially, and each child adds one image vertex from its parent's witness. Hence G_* contains $\text{im } \mu$ for every node μ of the search tree. If G_* admitted an admissible rooting η , start at the empty node. Whenever a node $\mu \subsetneq \eta$ has been reached, persistence gives some $\lambda \in \text{dom } \eta \setminus \text{dom } \mu$

with $\eta(\lambda) \in V(W_\mu) \setminus \text{im } \mu$; the corresponding one-label extension is a child of μ . Following these children reaches $\mu = \eta$, where persistence would demand a label in $\text{dom } \eta \setminus \text{dom } \eta$, a contradiction. Thus G_* is a bounded nonmember.

The admissibility predicate is decidable from the finite lists of edge types and relations. For each r -graph up to the bound (7.30b), exhaustively test all partial injective rootings and retain exactly the graphs having no admissible one. Every nonmember contains one of these retained graphs, while no member can contain one because $\mathcal{D}_f$ is subgraph-closed. The retained finite list is therefore an effective obstruction basis. $\square$

Call an admissible rooting *link-complete* if it is total, has at least one data vertex, contains every allowed base, balancing, and positive-correction edge, and contains all edges belonging to every state or coefficient declaration already present.

Link completion lets the later Lagrangian analysis use members in which all allowed edges are present.

Lemma 7.10. *Every member of $\mathcal{D}_f$ is a subgraph of a member of $\mathcal{D}_f$ admitting a link-complete rooting. The class $\mathcal{D}_f$ contains an edge.*

Proof. Start with an admissible partial rooting of the given member. Add one fresh vertex for each missing root label and extend the rooting to those vertices. If there is no data vertex, add one fresh data vertex. Insert every allowed base, balancing, and positive-correction edge, and insert all edges belonging to each state or coefficient declaration already present. No old data vertex is reclassified as a root. The uniqueness of the edge types shows that the enlarged rooting is still admissible: the new base and correction edges have their prescribed signatures, while completing an existing state or coefficient family introduces neither a new state nor a new coefficient parameter. Thus the resulting member is link-complete and contains the original graph.

Finally, the class contains an edge. Under a total rooting, take any allowed residual base edge, which exists for all sufficiently large parameters considered below. Alternatively, after adjoining two fresh data vertices one may take a single state edge. The second description makes clear why one data vertex alone is not being used to support a state declaration. $\square$

7.5 Lagrangian analysis and optimization

By Lemma 7.10, it is enough to analyze members with a link-complete rooting. We now pass from their rooted edge types to the exact scalar objective through the following chain:

$$\begin{aligned} & \text{edge-weight objective} \implies \text{part weights near uniform} \\ \implies & \text{uniform weights within the root blocks} \implies \text{complete data relation} \\ \implies & \text{scalar identity} \implies \text{zero gradient and concavity.} \end{aligned}$$

Summing the six edge types gives the objective in (7.32). Before balancing the root blocks, we show that every maximizing vector of part weights is close to the uniform vector; this avoids a circular argument. The balancing relations then force uniform weights inside each root block with fixed total weight. We complete the data-pair relation and obtain the scalar polynomial H_t and the class-Lagrangian identity. Finally, the correction degrees make the gradient vanish at the uniform vector, and a Hessian bound shows that this vector is the unique global maximizer when $t = 1/2$. The last part gives one lower bound on r for all structural and analytic conditions.

The formulas below continue the numbering of this section. None of the estimates depends on the input alphabet, coefficient denominator, or finite member of the class.

The objective and exact root balance.

Let $G \in \mathcal{D}_f$ have a link-complete rooting and let ω be a weighting. For a root block B , write $w^B = (w_a^B)_{a \in \Omega_N}$ and $x_B = \sum_a w_a^B$; let z be the total data mass. For a support on the parts E , write $x_E = \prod_{C \in E} x_C$, with $x_\star = z$.

For $(E, \{B, B'\}) \in \mathcal{H}$, put

$$\mathbf{M}_E(w) := (e_2(w^B) + e_2(w^{B'}) - x_B x_{B'}) x_{E \setminus \{B, B'\}}. \quad (7.31)$$

Direct summation over the edge families gives the following state and coefficient forms. They coincide with the corresponding one-sort construction of Li, Liu, and Liu [21]. If Δ_ϵ is the total product weight of state- ϵ data pairs and Σ_X is the parameter set present on a data d -set X , then

$$\begin{aligned}
p_G(\omega) = & e_r(x) - \sum_{E \in \mathcal{E}_{\text{pow}}} x_E + \sum_{(E, \partial E) \in \mathcal{H}} M_E(w) \\
& + \sum_{j=1}^4 Z_{S,j} \sum_{\epsilon=0}^1 \sum_{a \in \Omega_N} w_a^{A_j} w_{a+\epsilon}^{B_j} \Delta_\epsilon \\
& + Z_P \sum_{X \in \binom{V_{\text{data}}}{d}} \left(\prod_{z \in X} \omega_z \right) T_{\Sigma_X}(w^{U_1}, w^{U_2}, w^{U_3}) \\
& - \sum_{E \in \mathcal{C}_{\text{sub}}} x_{E \setminus \{C_1(E), C_2(E), C_3(E)\}} T_{T_N}(w^{C_1(E)}, w^{C_2(E)}, w^{C_3(E)}) \\
& + \sum_{E \in \mathcal{C}_{\text{add}}} x_{E \setminus \{D_1(E), D_2(E)\}} \sum_{ab \in E(X_{q_N})} w_a^{D_1(E)} w_b^{D_2(E)}. \tag{7.32}
\end{aligned}$$

Here $Z_{S,j}$ and Z_P are the products of the corresponding filler block totals. No edge is counted twice.

Put

$$\vartheta := \frac{1}{100rM}, \quad \ell_0 := \frac{1 - \vartheta}{M}, \quad u_0 := \frac{1 + \vartheta}{M}. \tag{7.33}$$

The first estimate localizes every maximizing vector of part weights before root balancing.

Lemma 7.11. *For all sufficiently large r , every maximizing weighting of every link-complete $G \in \mathcal{D}_f$ satisfies*

$$\ell_0 \leq x_i \leq u_0 \quad (i \in [M]). \tag{7.34}$$

The threshold is uniform in N , f , and G .

Proof. Discard the nonpositive power-deletion and unit-correction terms in (7.32). The sum of all remaining positive perturbations is at most

$$\varepsilon_{r,d}^+ := \left(4|\mathcal{H}| + 8 + \frac{d^d}{d!} + |\mathcal{C}_{\text{add}}| \right) r^{-r}. \tag{7.35}$$

Indeed, if positive integer exponents $a_1, \dots, a_t$ sum to r and $\sum_i x_i \leq 1$, weighted AM–GM gives

$$\prod_{i=1}^t x_i^{a_i} \leq \prod_{i=1}^t \left(\frac{a_i}{r} \right)^{a_i}.$$

For one balancing relation, discard the negative cross term and use $e_2(w^B) \leq x_B^2/2$; its two positive terms total at most $4r^{-r}$. Thus all balancing relations contribute at most $4|\mathcal{H}|r^{-r}$. In one value of j the two matching relations have total weight at most

$$x_{A_j} x_{B_j} Z_{S,j} (\Delta_0 + \Delta_1) \leq \frac{1}{2} x_{A_j} x_{B_j} Z_{S,j} z^2 \leq 2r^{-r},$$

so the four values of j contribute at most $8r^{-r}$. By Lemma 7.2 and $e_d((\omega_z)) \leq z^d/d!$, the coefficient term is at most $(d^d/d!)r^{-r}$. Finally, each positive-correction relation is contained in the complete bipartite relation, and hence contributes at most the corresponding squarefree monomial in the part weights r^{-r} . This proves (7.35), uniformly in N , f , and G .

At the uniform part-weight vector, balance every root block, put all data mass on one data vertex, and use no state or coefficient declaration. The power deletion costs at most $2M^{-5}/r!$; every balancing relation contributes $-M^{-r}/q_N$ and every negative unit correction contributes $-M^{-r}/Q_N$. Ignoring the nonnegative positive-correction terms, the test value is therefore at least

$$B_{M,r} - \frac{2M^{-5}}{r!} - (|\mathcal{H}| + |\mathcal{C}_{\text{sub}}|)M^{-r}, \quad B_{M,r} := \binom{M}{r} M^{-r}, \tag{7.36}$$

where we used $q_N, Q_N \geq 1$. For all sufficiently large r such that

$$r!\varepsilon_{r,d}^+ \leq M^{-5}, \quad r!(|\mathcal{H}| + |\mathcal{C}_{\text{sub}}|)M^{-r} \leq M^{-5}, \quad (7.36a)$$

a maximizing vector of part weights has

$$e_r(x) \geq B_{M,r} - \frac{4M^{-5}}{r!}.$$

Maclaurin's inequality gives, with $V(x) = \sum_i (x_i - M^{-1})^2$,

$$\frac{e_r(x)}{B_{M,r}} \leq \left(1 - \frac{M}{M-1}V(x)\right)^{r/2} \leq 1 - \frac{M}{M-1}V(x). \quad (7.37)$$

Since $A_{M,r} = r!B_{M,r} > 1/2$ and the second inequality in (7.37) is linear in $V(x)$, comparison gives $V(x) \leq 8M^{-5}$. The explicit condition $M \geq 80000r^2$ implies

$$|x_i - M^{-1}| \leq \sqrt{V(x)} \leq \frac{1}{100rM^2} = \frac{\vartheta}{M},$$

which proves (7.34). The explicit inequalities below verify all three displayed conditions uniformly. $\square$

Once localized, the balancing relations force exact uniformity within every root block.

Lemma 7.12. *For all sufficiently large r , the restriction of (7.32) to the product of the fixed-total root-block simplices is strictly concave whenever the vector of part weights lies in (7.34). Its unique maximizer is uniform inside every root block.*

Proof. On the zero-sum tangent space of a distinguished root block, one balancing relation contributes $-x_{E \setminus \partial E} \text{Id}$. Since every root block is distinguished K times, the total negative diagonal is at most $-K\ell_0^{r-2} \text{Id}$.

All remaining block-row contributions are recorded in the following table; in particular, no alphabet-size factor is suppressed.

term	block-row bound
state and coefficient terms	$2u_0^{r-2}$
negative unit corrections	$2b_0u_0^{r-2}$
positive spectral corrections	$T_+D_{\text{sp}}u_0^{r-2}/16$

(7.37a)

We justify each row. In one value of j , the mixed Hessian between A_j and B_j is a nonnegative linear combination of two shift permutation matrices. Since the state-0 and state-1 data pairs are disjoint subsets of all unordered data pairs, $\Delta_0 + \Delta_1 \leq e_2((\omega_z)_{z \in V_{\text{data}}}) \leq z^2/2$. Its norm is therefore at most

$$Z_{S,j}(\Delta_0 + \Delta_1) \leq \frac{1}{2}u_0^{r-4}z^2 \leq \frac{1}{2}u_0^{r-2}.$$

For a translation block, each of its two partners contributes, after summing over the data d -sets, at most

$$\frac{z^d}{d!}u_0^{r-d-2} \leq \frac{u_0^{r-2}}{d!}$$

by Lemma 7.2; since $d \geq 2$, the sum of the two partner norms is at most u_0^{r-2} . Thus the first row of (7.37a) is a safe common bound. In a negative unit correction, only the three designated blocks vary internally. Each designated block has two partners, each mixed Hessian has norm at most u_0^{r-2} , and the designation load is at most b_0 , giving the second row. Finally, a root block is designated on at most T_+ positive corrections; on the zero-sum tangent spaces the nontrivial mixed norm of each spectral relation is at most $D_{\text{sp}}/16$, which gives the third row. These estimates contain no hidden factor depending on q_N . By (7.19), (7.23), and (7.25),

$$2 + 2b_0 + \frac{T_+D_{\text{sp}}}{16} \leq \frac{K}{8}. \quad (7.38)$$

Also $(u_0/\ell_0)^{r-2} \leq 2$ under the inequalities used here. Hence the perturbative block-row norm is at most $K\ell_0^{r-2}/4$, and block Gershgorin gives the quantitative bound

$$\nabla^2 p_G|_{\prod_B \tau_B} \preceq -\frac{3K}{4}\ell_0^{r-2} \text{Id}, \quad (7.38a)$$

where $\mathcal{T}_B$ is the *zero-sum tangent space* in block B .

The point with uniform internal weights is a critical point. The shift and translation relations are regular in every coordinate, and $\mathbf{X}_{q_N}$ is regular on both sides; e_2 also has constant gradient at the uniform point on a fixed-total simplex. Strict concavity makes this critical point the unique maximizer. $\square$

Completing the data relation and the scalar identity.

At balanced root weights define

$$\text{Bal}(x) := \sum_{(E, \{B, B'\}) \in \mathcal{H}} \left[\frac{q_N - 1}{2q_N} (x_B^2 + x_{B'}^2) - x_B x_{B'} \right] x_{E \setminus \{B, B'\}}, \quad (7.39)$$

$$\text{State}(x) := \frac{z^2}{2q_N} \sum_{j=1}^4 x_{A_j} x_{B_j} Z_{S,j}, \quad (7.40)$$

$$\text{Score}(x) := \frac{1}{q_N d!} x_{U_1} x_{U_2} x_{U_3} Z_P z^d. \quad (7.41)$$

For an r -graph J on the set of parts, write $P_J(x) = \sum_{E \in E(J)} x_E$. Put

$$\begin{aligned} H_t(x) := & e_r(x) - P_{\mathcal{E}_{\text{pow}}}(x) + \text{Bal}(x) + \text{State}(x) + t \text{Score}(x) \\ & - \frac{1}{Q_N} P_{\mathcal{C}_{\text{sub}}}(x) + \frac{D_{\text{sp}}}{q_N} P_{\mathcal{C}_{\text{add}}}(x). \end{aligned} \quad (7.42)$$

We first show that incomplete data relations can be completed without lowering the value.

Lemma 7.13. *Let G be link-complete and let ω maximize its Lagrangian. After balancing the roots, one may declare every undeclared positive data pair to have state 0, delete every coefficient orbit made incompatible by this completion, and insert every permitted coefficient orbit, obtaining a link-complete member $G^{\text{rep}} \in \mathcal{D}_f$ with $p_{G^{\text{rep}}}(\omega) \geq p_G(\omega)$.*

Proof. Let $\mathcal{P}$ be the set of undeclared data pairs zw with $\omega_z \omega_w > 0$. Add the four complete state-0 families for every pair in $\mathcal{P}$. For a fixed $zw \in \mathcal{P}$, this gains at least

$$\frac{4\ell_0^{r-2}}{q_N} \omega_z \omega_w. \quad (7.42a)$$

At balanced root weights, the union of all coefficient parameters on one data d -set has three-block factor at most $1/q_N$. Moreover,

$$\sum_{\substack{X \in \binom{V_{\text{data}}}{d} \\ zw \subseteq X}} \prod_{v \in X} \omega_v \leq \frac{z^{d-2}}{(d-2)!} \omega_z \omega_w.$$

It follows that the total weight of all coefficient orbits on data d -sets containing zw is at most

$$\frac{u_0^{r-2}}{q_N (d-2)!} \omega_z \omega_w. \quad (7.42b)$$

Now delete precisely the coefficient orbits that violate (S2) after the state completion. Every deleted orbit of positive weight lies on a data d -set containing a pair from $\mathcal{P}$; otherwise all its pairs were already declared before the completion, and its presence would already have forced (S2). Charge each deleted orbit once to one such pair. For a fixed pair, the charged loss is bounded by (7.42b). Since $(u_0/\ell_0)^{r-2} < 2$ and $(d-2)! \geq 1$, this loss is strictly less than half of the gain in (7.42a). Summing over $\mathcal{P}$ therefore proves that the simultaneous state completion and all required deletions do not decrease the value. Orbits of zero weight cause no loss, and inserting all newly permitted coefficient orbits can only increase the value. Finally, declare every still undeclared data pair of zero product weight to have state 0. Each newly added state edge then has zero weight; every coefficient orbit made incompatible at this last step contains such a pair and also has zero weight, so deleting those orbits is cost-free. No state conflict is introduced, (S2) holds by construction, and inserting all edges in these families makes the resulting member link-complete with a complete data-pair relation. $\square$

This completion reduces the class Lagrangian to a scalar optimization problem.

Proposition 7.14. *For every rational input f ,*

$$\Lambda(\mathcal{D}_f) = \max_{x \in \Delta_M} H_{\text{Val}_d(f)}(x). \quad (7.43)$$

Proof. For the upper bound, extend to a link-complete member, choose a maximizing weighting, and apply Lemmas 7.11 to 7.13. The completed data relation is a finite weighted graph. On a data d -set of type F , (7.3) and $|S_F| = a_F q_N$ give balanced root factor

$$\frac{a_F}{q_N} x_{U_1} x_{U_2} x_{U_3} Z_P.$$

After normalizing the data weights, Lemma 6.1 bounds the coefficient contribution by $\text{Val}_d(f) \text{Score}(x)$. The state contribution is at most $\text{State}(x)$ because the total product weight of data pairs is at most $z^2/2$. All base, balancing, and correction terms are exactly those in (7.42).

Conversely, fix $x \in \Delta_M$ and take the uniform finite graphs from Lemma 6.1. Give their data vertices total mass z , declare all pairs by their graph states, and insert exactly the permitted coefficient orbits. Insert every allowed base, balancing, and positive-correction edge and omit precisely the prescribed deletions. Balance each root block. The state pair weight tends to $z^2/2$ and the coefficient term tends to $\text{Val}_d(f) \text{Score}(x)$. These finite members of $\mathcal{D}_f$ have values tending to $H_{\text{Val}_d(f)}(x)$. $\square$

Integral gradient cancellation and the rational reference value.

Let $R_t(x) := \text{Bal}(x) + \text{State}(x) + t \text{Score}(x)$. At p define

$$g_i := Q_N M^{r-1} \partial_i R_\tau(p). \quad (7.44)$$

The correction design uses the following integrality and divisibility facts.

Lemma 7.15. *Every g_i is an integer, $r \mid \sum_i g_i$, and*

$$g_\star = h_\star, \quad g_B = -c_d R + h_B. \quad (7.45)$$

Proof. The polynomial $Q_N R_\tau$ is homogeneous of degree r with integer coefficients. Indeed, the balancing coefficients become $c_d(q_N - 1)/2$ and $-c_d q_N$, the state coefficient becomes $c_d/2 = d!$, and, since $\tau = 1/2$, the slope coefficient becomes $c_d/(2d!) = 1$. Hence evaluation of a derivative at p , followed by multiplication by M^{r-1} , gives an integer.

A balancing relation contributes $-c_d$ to the normalized derivative at every part coordinate in its support. Each root coordinate belongs to exactly R balancing supports, whereas $\star$ belongs to none. The four values of j contribute $8d!$ at $\star$ and $d!$ at each state block or state filler. The slope term contributes d at $\star$ and 1 at each translation block or coefficient filler. This proves (7.45). Moreover, using

$$h_\star + H_{\text{special}} = (8d! + d) + 4(r - 2)d! + (r - d) = r(4d! + 1)$$

and $c_d R = r(c_d K/2)$, we obtain the explicit divisibility identity

$$\sum_i g_i = h_\star + H_{\text{special}} - (M - 1)c_d R = r \left(4d! + 1 - (M - 1) \frac{c_d K}{2} \right).$$

Thus $r \mid \sum_i g_i$. This conclusion comes from the displayed derivative calculation, not from Euler's identity. Euler's identity is used below only after the correction terms cancel every derivative at p . $\square$

By Lemma 7.7,

$$d_{\mathcal{C}_{\text{sub}}}(i) = g_i + L, \quad d_{\mathcal{C}_{\text{add}}}(i) = d_+, \quad c_d D_{\text{sp}} d_+ = L. \quad (7.46)$$

Consequently,

$$Q_N M^{r-1} \partial_i \left(R_\tau - \frac{1}{Q_N} P_{\mathcal{C}_{\text{sub}}} + \frac{D_{\text{sp}}}{q_N} P_{\mathcal{C}_{\text{add}}} \right) (p) = g_i - (g_i + L) + L = 0. \quad (7.47)$$

The bracket is homogeneous of degree r , so Euler's identity also gives

$$R_\tau(p) - \frac{1}{Q_N} P_{\mathcal{C}_{\text{sub}}}(p) + \frac{D_{\text{sp}}}{q_N} P_{\mathcal{C}_{\text{add}}}(p) = 0. \quad (7.48)$$

By regularity of $\mathcal{E}_{\text{pow}}$, the gradient of $e_r - P_{\mathcal{E}_{\text{pow}}}$ is constant at p . Thus p is a constrained critical point of H_τ , and (7.18), (7.48) give

$$r!H_\tau(p) = \rho_{r,h} = \theta_{r,h}^h. \quad (7.49)$$

It remains to show that this critical point is the unique global maximizer.

Lemma 7.16. *For all sufficiently large r , the point p is the unique global maximizer of H_τ on Δ_M .*

Proof. The argument of Lemma 7.11, now applied to the scalar objective, places every maximizer in the neighborhood defined by (7.34). At p , the tangent Hessian of e_r is

$$-\beta_{M,r} \text{Id}, \quad \beta_{M,r} := \binom{M-2}{r-2} M^{2-r}.$$

For $i \neq j$, the (i, j) entry of the Hessian of the complete base is $e_{r-2}(x_{i,j}^\wedge)$, whereas at p it is $\beta_{M,r}$. Since every coordinate satisfying (7.34) lies between $(1 - \vartheta)/M$ and $(1 + \vartheta)/M$, and $(r-2)\vartheta \leq 1/2$ under the inequalities used here,

$$\left| e_{r-2}(x_{i,j}^\wedge) - \beta_{M,r} \right| \leq 2(r-2)\vartheta \beta_{M,r}. \quad (7.49a)$$

The diagonal Hessian entries vanish. Summing (7.49a) over the at most $M-1$ off-diagonal entries in a row therefore gives

$$\|\nabla^2 e_r(x) - \nabla^2 e_r(p)\|_{\text{op}} \leq 2rM\vartheta \beta_{M,r} = \frac{1}{50} \beta_{M,r}. \quad (7.49b)$$

Thus the same bound holds after restriction to the tangent space.

For the regular deletion,

$$\|\nabla^2 P_{\mathcal{E}_{\text{pow}}}\|_{\text{op}} \leq (r-1)D_{\text{del}}u_0^{r-2} \leq 8M^{-4}\beta_{M,r} \quad (7.50)$$

for large r , by (7.15). Every remaining term in the part weights has maximum support degree at most r . To retain the exact multiplicities, write

$$\widehat{R}_\tau(x) := \text{Bal}(x) + \text{State}(x) + \tau \text{Score}(x) - \frac{1}{Q_N} P_{\mathcal{C}_{\text{sub}}}(x) + \frac{D_{\text{sp}}}{q_N} P_{\mathcal{C}_{\text{add}}}(x) = \sum_{\nu} c_{\nu} x^{a(\nu)}$$

and define the finite *weighted-incidence parameter*

$$\mathfrak{W}_{r,d} := \max_{i \in [M]} \sum_{\nu} |c_{\nu}| a_i(\nu). \quad (7.50a)$$

It is computed directly from the already constructed support and relation tables. Using $q_N \geq D_{\text{sp}}$, the construction gives the conservative alphabet-independent estimate

$$\mathfrak{W}_{r,d} \leq 6R + 4 + d + (b_0 + c_d R + h_{\star}) + d_{+}. \quad (7.50b)$$

Indeed, the five summands bound respectively the balancing, state, slope, unit-correction, and positive-correction incidences.

For one monomial $c_{\nu} x^{a(\nu)}$, the absolute row sum of its Hessian in row i is at most

$$|c_{\nu}| a_i(\nu) (r-1) u_0^{r-2}.$$

Consequently

$$\|\nabla^2 \widehat{R}_\tau(x)\|_{\text{op}} \leq (r-1) \mathfrak{W}_{r,d} u_0^{r-2}. \quad (7.50c)$$

For all sufficiently large r such that

$$\prod_{j=2}^{r-1} \left(1 - \frac{j}{M}\right) \geq \frac{1}{2}, \quad (1 + \vartheta)^{r-2} \leq 2,$$

we have $\beta_{M,r} \geq 1/(2(r-2)!)$ and $u_0^{r-2} \leq 2M^{-(r-2)}$. Thus

$$\frac{\|\nabla^2 \hat{R}_\tau(x)\|_{\text{op}}}{\beta_{M,r}} \leq \frac{4(r-1)! \mathfrak{W}_{r,d}}{M^{r-2}}. \quad (7.50d)$$

The explicit inequalities below make the right side at most $1/10$ and also makes $8M^{-4} \leq 1/10$ in (7.50). Hence the total perturbation is less than $(1/50 + 1/10 + 1/10)\beta_{M,r} < \beta_{M,r}/4$. Therefore

$$\nabla^2 H_\tau|_{T\Delta_M} \preceq -\frac{3}{4}\beta_{M,r} \text{Id}$$

throughout this neighborhood. The constrained critical point p is its unique maximizer there, and localization makes it the unique global maximizer. $\square$

We now choose one lower bound on r that satisfies all structural and analytic requirements.

Lemma 7.17. *For fixed $d \geq 2$ and $h \geq 1$, there is an integer $R_{\text{com}}(d, h)$ such that every assertion in Lemmas 7.4 to 7.7, 7.11, 7.12 and 7.16 holds whenever $r \geq R_{\text{com}}(d, h)$. This bound does not depend on the alphabet, the coefficient vector or denominator of f , or the finite member of $\mathcal{D}_f$. One finite-search procedure constructs all required finite objects and terminates for every r in this range; it need not be given $R_{\text{com}}(d, h)$. The proof below gives finite lists of structural and analytic conditions that can be checked exactly for each fixed r .*

Proof. Recall $M = r^{20(h+1)}$ and the exact counts

$$|\mathcal{H}| = \frac{(M-1)K}{2}, \quad |\mathcal{C}_{\text{add}}| = \frac{Md_+}{r}, \quad |\mathcal{C}_{\text{sub}}| = \frac{Mb_0 + S_c}{r}. \quad (7.51)$$

For the structural part, retain the notation L_v from Lemma 7.6, and put

$$L_{\max} := \max_{v \in [K/2]} L_v = L_{K/2}.$$

After the arithmetic choice of u , and hence of D_{del} , the following finite list is sufficient for every structural step:

$$r \geq \max\{4, d+3\}, \quad M \geq 4r, \quad P \geq s_{\text{special}} = 5r - d - 8, \quad (7.52a)$$

$$P > 2L_{\max}, \quad (7.52b)$$

$$T_{M,r} - \binom{5r}{2} \binom{M-2}{r-2} - C_d^{\text{res}}(M+r^2) \geq D_{\text{del}}, \quad (7.52c)$$

$$b_0 + d_+ \leq C_d^{\text{res}} r \leq D_{\text{del}}, \quad (7.52d)$$

$$\frac{M}{r} - 5r \geq \frac{(r-1)S_c}{r}, \quad (7.52e)$$

$$\frac{S_c(R+b_0+d_+)}{\binom{(r-1)S_c}{r-1}} < 1, \quad (7.52f)$$

$$\left\lceil \frac{2d_+}{r-1} \right\rceil \leq \min\{d_+, T_+\}, \quad (7.52g)$$

$$b_0 + S_c + \frac{PK}{2} \leq C_d^{\text{res}}(M+r^2). \quad (7.52h)$$

Conditions (7.52a) and (7.52b) make all required parts available, make the wrap-around gap $P - L_v$ uniquely largest in every balancing support, and separate the two oriented residual-offset ranges in the proof of Lemma 7.6. Conditions (7.52c) and (7.52d) give the exact worst-case reservoir inequality (7.16b) and give the required number of factors reserved in Steps 1 and 4 of Lemma 7.7. Condition (7.52e) leaves enough padding-only edges in the first baseline factor, while (7.52f) is precisely the switching union bound (7.26a). Finally, (7.52g) gives the capacity used in the integral b -matching, and (7.52h) bounds the exclusions in the last reservoir call.

All comparisons in (7.52a)–(7.52h) are decidable by integer arithmetic once u has been chosen, and they hold for every sufficiently large r . Indeed, $L_{\max}, s_{\text{special}}, S_c, R, d_+$ are $O_d(r)$, whereas $P = r^{20(h+1)} - 1$. The lower bound in (7.15) gives

$$D_{\text{del}} \geq \frac{M^{r-6}}{(r-1)!},$$

which eventually exceeds $C_d^{\text{res}}r$. Moreover,

$$\frac{\binom{5r}{2} \binom{M-2}{r-2}}{T_{M,r}} = \binom{5r}{2} \frac{r-1}{M-1},$$

$$\frac{C_d^{\text{res}}(M+r^2)}{T_{M,r}} \leq C_d^{\text{res}}(M+r^2) \left(\frac{2r}{M}\right)^{r-1}, \quad \frac{D_{\text{del}}}{T_{M,r}} \leq 4M^{-5}.$$

The sum of these three bounds tends to zero, proving (7.52c) for all sufficiently large r . Also $\binom{(r-1)S_c}{r-1} \geq S_c^{r-1}$, so the ratio in (7.52f) tends to zero. The remaining assertions follow by direct polynomial comparison from the displayed definitions.

The algorithm does not need this lower bound on r . It searches for u by exact comparison of h th powers, enumerates a Baranyai factorization of the finite complete r -graph, constructs the balancing relations from their displayed formula, scans the fixed factorization for the reservoir choices, enumerates the finitely many labeled partitions of W , and solves the finite integral max-flow instance. The spectral-relation search of Lemma 7.3 terminates independently of r . On every r satisfying (7.52a)–(7.52h) and the arithmetic conditions below, each search therefore finds a witness and halts.

It remains to record the analytic estimates; this is the part for which uniformity in the input alphabet and denominator is essential. The localization, within-block, and part-weight arguments require only the following finite list of inequalities:

$$A_{M,r} > \frac{1}{2}, \quad M \geq 80000r^2, \quad (7.53)$$

$$r! \varepsilon_{r,d}^+ \leq M^{-5}, \quad r!(|\mathcal{H}| + |\mathcal{C}_{\text{sub}}|)M^{-r} \leq M^{-5}, \quad (7.54)$$

$$\left(\frac{u_0}{\ell_0}\right)^{r-2} \leq 2, \quad (r-2)\vartheta \leq \frac{1}{2}, \quad (7.55)$$

$$\prod_{j=2}^{r-1} \left(1 - \frac{j}{M}\right) \geq \frac{1}{2}, \quad (1+\vartheta)^{r-2} \leq 2, \quad (7.56)$$

$$8M^{-4} \leq \frac{1}{10}, \quad \frac{4(r-1)! \mathfrak{M}_{r,d}}{M^{r-2}} \leq \frac{1}{10}. \quad (7.57)$$

The arithmetic-power choice additionally uses

$$M > r(r-1), \quad r \geq 12h+2, \quad r^{9r+1} > h. \quad (7.58)$$

After substituting (7.51), (7.35), and (7.50b), every left side is an explicit expression in r with constants depending only on (d, h) . The elementary bounds

$$\frac{r!}{r^r} \leq 2^{-\lfloor r/2 \rfloor}, \quad (r-1)! \leq r^{r-1}, \quad \prod_j (1 - a_j) \geq 1 - \sum_j a_j \quad (0 \leq a_j \leq 1)$$

reduce (7.53)–(7.58) to finitely many inequalities of the forms

$$Cr^m M^{-a} \leq c, \quad Cr^m M^a 2^{-\lfloor r/2 \rfloor} \leq c, \quad Cr^m \left(\frac{2r}{M}\right)^{r-b} \leq c, \quad (7.59)$$

with known nonnegative integers C, m, a, b and positive rational c . Power comparison handles the first form. For the second, increase r until the inequality holds in both parity classes and its two-step ratio is at most $3/4$. For the third, use $M = r^{20(h+1)}$ and increase r until the negative exponent dominates m ; the same ratio test certifies monotone decrease. These procedures use only integer and rational arithmetic and halt. Their maximum is the claimed analytic threshold. None of (7.53)–(7.59) involves q_N . $\square$

The algorithmic meaning of this statement is the one fixed in the Introduction; the finite subroutines are summarized in [Section 9](#).

The preceding estimates give the analytic consequences of the construction.

Proposition 7.18. *Fix $d \geq 2$ and $h \geq 1$. For all sufficiently large r , uniformly in the rational input table f , the construction above has an explicitly defined polynomial family*

$$H_t : \Delta_M \longrightarrow \mathbb{R} \quad (0 \leq t \leq 1)$$

with the following properties.

(i) *If $v := \text{Val}_d(f)$, then*

$$\Lambda(\mathcal{D}_f) = \max_{x \in \Delta_M} H_v(x).$$

(ii) *At $\tau = 1/2$, the uniform part-weight vector $p = (M^{-1}, \dots, M^{-1})$ is the unique maximizer of H_τ , and*

$$r!H_\tau(p) = \rho_{r,h}.$$

(iii) *If $v = \text{Val}_d(f) \geq \tau$, then*

$$r!\Lambda(\mathcal{D}_f) \geq \rho_{r,h}, \quad r!\Lambda(\mathcal{D}_f) = \rho_{r,h} \iff v = \tau.$$

Consequently, strict inequality holds whenever $v > \tau$.

The common lower bound on r depends only on (d, h) , and the associated finite searches terminate for every r in this range.

Proof. The scalar identity is Proposition 7.14. Equations (7.47)–(7.49) give the value and show that the gradient vanishes at p , and Lemma 7.16 proves uniqueness at $t = \tau$. For $t > \tau$, evaluation at p increases by the positive slope term. Lemma 7.17 gives a common lower bound on r . $\square$

We now combine the finite construction, the analytic estimates, and the exact ordinary transfer.

Proof of Theorem 7.1. Choose $r_{\text{pow}}(d, h)$ large enough that the structural constructions above and Proposition 7.18 apply. Lemma 7.9 gives a finite obstruction family $\mathcal{A}_f$ for $\mathcal{D}_f$. Let

$$\mathcal{F}_f := \bigcup_{A \in \mathcal{A}_f} \text{Ext}(A).$$

By Theorem 2.4,

$$\pi(\mathcal{F}_f) = r!\Lambda(\mathcal{D}_f).$$

Proposition 7.18 gives, whenever $\text{Val}_d(f) \geq 1/2$,

$$\pi(\mathcal{F}_f) \geq \rho_{r,h}, \quad \pi(\mathcal{F}_f) = \rho_{r,h} \iff \text{Val}_d(f) = \frac{1}{2}.$$

The same proposition gives strict inequality exactly when $\text{Val}_d(f) > 1/2$.

The output is obtained by the finite subroutines listed in [Section 9](#); none of them queries the real number $\text{Val}_d(f)$. Moreover, the reference value $\rho_{r,h}$ depends only on (r, h) and is therefore independent of both the coefficient vector and its common denominator. $\square$

8 Proofs of the main results

We now combine the two ordinary results with the vertex-degree identity and the exact higher-degree transfers. Let $r_{\text{pow}}(d, h)$ be a threshold such that [Theorem 7.1](#) applies whenever $r \geq r_{\text{pow}}(d, h)$. To keep the logical dependencies separate, first choose parameters for undecidability alone:

$$\begin{aligned} q_{\text{und}} &\geq \max\{r_{\text{pow}}(d_0, 1), 2\}, & s_{\text{und}} &:= \max\{s_0(q_{\text{und}}), 3\}, \\ B_{\text{und}} &:= \max\left\{s_{\text{und}} - 1, \max_{1 \leq h < s_{\text{und}}} r_{\text{pow}}(d_0, h)\right\}, & r_0^{\text{und}} &:= B_{\text{und}} + s_{\text{und}}. \end{aligned} \tag{8.1}$$

Independently, choose parameters for transcendence:

$$\begin{aligned} q_{\text{tr}}^* &\geq \max\{r_{\text{tr}}, 2\}, & s_{\text{tr}}^* &:= \max\{s_0(q_{\text{tr}}^*), 3\}, \\ B_{\text{tr}}^* &:= \max\{s_{\text{tr}}^* - 1, r_{\text{tr}}\}, & r_0^{\text{tr}} &:= B_{\text{tr}}^* + s_{\text{tr}}^*, & r_0 &:= \max\{r_0^{\text{und}}, r_0^{\text{tr}}\}. \end{aligned} \quad (8.2)$$

We do not optimize these constants. The two constructions use their chosen parameters as fixed internal constants, not as additional inputs. In particular, the undecidability construction depends only on [Theorem 6.3](#) and the internal construction and transfer bounds; it does not use r_{tr} or [Theorem 6.4](#). The endpoints in (8.1)–(8.2) serve only as lower bounds for the stated range and are not inputs to the partial algorithms in [Theorems 1.1](#) and [1.2](#).

For $\diamond \in \{\text{und}, \text{tr}\}$, write $(q^\diamond, s^\diamond, B^\diamond, r_0^\diamond)$ for the corresponding quadruple in (8.1) or (8.2); thus, for example, $q^{\text{tr}} = q_{\text{tr}}^*$. Fix $r \geq r_0$, $1 \leq \ell \leq r - 1$, and put $s := r - \ell$. The endpoint $\ell = 1$ is handled directly by [Lemma 2.1](#). For either construction, if $\ell \geq 2$ and $s < s^\diamond$, then

$$\ell = r - s \geq r_0^\diamond - (s^\diamond - 1) > B^\diamond \geq s^\diamond - 1, \quad (8.3)$$

whereas $\ell \geq 2$ and $s \geq s^\diamond$ imply $s > q^\diamond$ and $\kappa_{q^\diamond, s} \leq \gamma_s$. Thus every pair (r, ℓ) in the asserted range belongs to exactly one of the following three cases for each construction. The power column applies only when $\diamond = \text{und}$.

Case for $\diamond$	Ordinary-family uniformity	Power h	Transfer and hypotheses
$\ell = 1$	uniformity r	$h = 1$	vertex-degree identity; the required lower bound follows from $r \geq r_0^\diamond$
$\ell \geq 2, 1 \leq s < s^\diamond$	uniformity ℓ	$h = s$	clean-set transfer; $s < \ell$ and the required lower bound follows from $\ell > B^\diamond$
$\ell \geq 2, s \geq s^\diamond$	fixed uniformity $q^\diamond$	$h = 1$	suspension $q^\diamond \rightarrow s$ followed by the two-marker transfer; $q^\diamond < s$ and $\kappa_{q^\diamond, s} \leq \gamma_s$

Each proof below follows the relevant row from left to right: construct the ordinary Turán input family in the indicated uniformity and then apply the displayed identity or exact transfer. We use $\diamond = \text{tr}$ in the next subsection and $\diamond = \text{und}$ thereafter. The transcendence argument does not use the rational-power construction. The bounded- s case includes $s = 1$ and therefore covers the codegree endpoint $\ell = r - 1$.

8.1 Transcendental densities

Proof of [Theorem 1.1](#). If $\ell = 1$, apply [Theorem 6.4](#) in ordinary uniformity r and set $\mathcal{F}_{r,1} := \mathcal{P}_r^{\text{tr}}$. Then

$$\pi_1^{(r)}(\mathcal{F}_{r,1}) = \pi(\mathcal{P}_r^{\text{tr}}) = \beta_r^{\text{tr}},$$

which is transcendental by [Lemma 2.1](#) and [Theorem 6.4](#).

Suppose next that $\ell \geq 2$ and $1 \leq s < s_{\text{tr}}^*$. Apply [Theorem 6.4](#) in ordinary uniformity ℓ , and write $\beta := \pi(\mathcal{P}_\ell^{\text{tr}})$. This number is transcendental and positive. The clean-set transfer gives the finite family $\mathcal{F}_{r,\ell} := \mathcal{U}_{\ell,s}(\mathcal{P}_\ell^{\text{tr}})$ with

$$\pi_\ell^{(r)}(\mathcal{F}_{r,\ell}) = \Theta_s(\beta).$$

By [Lemma 5.6](#), the right-hand side is transcendental.

Finally, suppose that $\ell \geq 2$ and $s \geq s_{\text{tr}}^*$. Use the fixed class $\mathcal{D}_{q_{\text{tr}}^*}^{\text{tr}}$ from [Theorem 6.4](#), whose normalized Lagrangian $\beta_{q_{\text{tr}}^*}^{\text{tr}} = q_{\text{tr}}^*! \Lambda(\mathcal{D}_{q_{\text{tr}}^*}^{\text{tr}})$ is transcendental. The transfer in (5.2) produces the finite family $\mathcal{F}_{r,\ell} := \mathcal{M}_{q_{\text{tr}}^*, s, \ell}(\mathcal{D}_{q_{\text{tr}}^*}^{\text{tr}})$ and gives

$$\pi_\ell^{(r)}(\mathcal{F}_{r,\ell}) = \Gamma_{q_{\text{tr}}^*, s}(\beta_{q_{\text{tr}}^*}^{\text{tr}}).$$

This value is again transcendental by [Lemma 5.6](#). All families used in the three cases are obtained from finite edge lists. Together, the three cases give the single partial algorithm asserted in [Theorem 1.1](#). $\square$

8.2 Rational threshold constructions

The next proposition collects the three constructions used in the undecidability reduction.

Proposition 8.1. *There are partial computable maps $\Xi(r, \ell)$ and $\Psi(r, \ell, a)$ such that, for every $r \geq r_0$, every $1 \leq \ell \leq r - 1$, and every $a \in \mathbb{N}$, the value*

$$\xi_{r, \ell} := \Xi(r, \ell)$$

is rational, $\Psi(r, \ell, a)$ is a finite family $\mathcal{F}_a$ of r -graphs, and

$$\pi_\ell^{(r)}(\mathcal{F}_a) \geq \xi_{r, \ell}, \quad \begin{cases} a \notin \mathfrak{U} & \iff \pi_\ell^{(r)}(\mathcal{F}_a) = \xi_{r, \ell}, \\ a \in \mathfrak{U} & \iff \pi_\ell^{(r)}(\mathcal{F}_a) > \xi_{r, \ell}. \end{cases}$$

For fixed (r, ℓ) , the threshold is independent of a .

Proof. We treat the three cases in the table above.

The vertex-degree endpoint.

Assume $\ell = 1$. Since $r \geq r_0$, [Theorem 7.1](#) applies in ordinary uniformity r with desired power $h = 1$. Applying it to the constant table g° computes the rational reference value

$$\xi_{r, 1} := \rho_{r, 1} = \theta_{r, 1} \in \mathbb{Q} \cap (0, 1).$$

On input $a \in \mathbb{N}$, apply the same construction to the score table g_a from [Theorem 6.3](#), and let $\mathcal{F}_a$ be its finite pair-covering extension family. By [Theorem 7.1](#) and [Lemma 2.1](#),

$$\pi_1^{(r)}(\mathcal{F}_a) = \pi(\mathcal{F}_a) \geq \xi_{r, 1}, \quad \begin{cases} a \notin \mathfrak{U} & \iff \pi_1^{(r)}(\mathcal{F}_a) = \xi_{r, 1}, \\ a \in \mathfrak{U} & \iff \pi_1^{(r)}(\mathcal{F}_a) > \xi_{r, 1}. \end{cases}$$

Both the family and the threshold are obtained by the finite construction subroutines, and the threshold is independent of a .

Bounded $s = r - \ell$.

Assume $\ell \geq 2$ and $1 \leq s < s_{\text{und}}$. By (8.3), [Theorem 7.1](#) applies in ordinary uniformity ℓ with desired power $h = s$. Applying it to the fixed table g° with denominator 2 gives the rational reference value $\rho_{\ell, s} = \theta_{\ell, s}^s$, which depends only on (ℓ, s) .

On input $a \in \mathbb{N}$, apply the same construction to the score table g_a from [Theorem 6.3](#). Let $\mathcal{A}_a$ be the resulting obstruction basis, let $\mathcal{D}_a := \text{Forb}(\mathcal{A}_a)$, and let $\mathcal{P}_a := \bigcup_{A \in \mathcal{A}_a} \text{Ext}(A)$ be the pair-covering extension family. Writing $\alpha_a := \pi(\mathcal{P}_a) = \ell! \Lambda(\mathcal{D}_a)$, we have

$$\alpha_a \geq \rho_{\ell, s}, \quad \begin{cases} a \notin \mathfrak{U} & \iff \alpha_a = \rho_{\ell, s}, \\ a \in \mathfrak{U} & \iff \alpha_a > \rho_{\ell, s}. \end{cases}$$

Apply the clean-set transfer and set $\mathcal{F}_a := \mathcal{U}_{\ell, s}(\mathcal{P}_a)$. Then

$$\pi_\ell^{(r)}(\mathcal{F}_a) = \Theta_s(\alpha_a).$$

Since Θ_s is strictly increasing, the rational number

$$\xi_{r, \ell} := \Theta_s(\rho_{\ell, s}) = \left(\frac{\theta_{\ell, s}}{1 + \theta_{\ell, s}} \right)^s \tag{8.4}$$

satisfies

$$\pi_\ell^{(r)}(\mathcal{F}_a) \geq \xi_{r, \ell}, \quad \begin{cases} a \notin \mathfrak{U} & \iff \pi_\ell^{(r)}(\mathcal{F}_a) = \xi_{r, \ell}, \\ a \in \mathfrak{U} & \iff \pi_\ell^{(r)}(\mathcal{F}_a) > \xi_{r, \ell}. \end{cases} \tag{8.5}$$

For $s = 1$, this reads $\xi_{r, r-1} = \theta_{r-1, 1}/(1 + \theta_{r-1, 1})$.

Large $s = r - \ell$.

Assume $\ell \geq 2$ and $s \geq s_{\text{und}}$. Apply [Theorem 7.1](#) in the fixed ordinary uniformity q_{und} with $h = 1$, and denote its rational reference value by $\rho_{\text{und}} := \rho_{q_{\text{und}}, 1}$. Run the construction on g_a and, with the same

even denominator, on the constant score table $g^\circ = \frac{1}{2}\mathbf{1}_{d_0}$. The two runs return obstruction bases $\mathcal{A}_a, \mathcal{A}_a^{\text{ref}}$ and the corresponding classes $\mathcal{D}_a := \text{Forb}(\mathcal{A}_a)$ and $\mathcal{D}_a^{\text{ref}} := \text{Forb}(\mathcal{A}_a^{\text{ref}})$.

Put $\beta_a := q_{\text{und}}! \Lambda(\mathcal{D}_a)$ and $\beta_a^{\text{ref}} := q_{\text{und}}! \Lambda(\mathcal{D}_a^{\text{ref}}) = \rho_{\text{und}}$. Then

$$\beta_a \geq \rho_{\text{und}}, \quad \begin{cases} a \notin \mathfrak{U} & \iff \beta_a = \rho_{\text{und}}, \\ a \in \mathfrak{U} & \iff \beta_a > \rho_{\text{und}}. \end{cases}$$

Also $\beta_a \leq 1$, since $q_{\text{und}}! p_G(z) \leq (\sum_v z_v)^{q_{\text{und}}} = 1$ for every probability weighting z .

Using their finite obstruction bases, suspend both classes from uniformity q_{und} to s and denote the resulting finite one-marker families by $\mathcal{Q}_a^{\text{ref}} := \mathcal{A}_s(\mathcal{D}_a^{\text{ref}})$ and $\mathcal{Q}_a := \mathcal{A}_s(\mathcal{D}_a)$. By (4.2), their ordinary densities are

$$a_0 := \kappa_{q_{\text{und}}, s} \rho_{\text{und}}, \quad b_a := \kappa_{q_{\text{und}}, s} \beta_a.$$

Thus $0 < a_0 \leq b_a \leq \kappa_{q_{\text{und}}, s} \leq \gamma_s$. Apply the two-marker transfer with the constant family first and the variable family second, and set $\mathcal{F}_a := \mathcal{B}_{\ell, s}(\mathcal{Q}_a^{\text{ref}}, \mathcal{Q}_a)$. Then

$$\pi_\ell^{(r)}(\mathcal{F}_a) = \Omega_s(a_0, b_a).$$

For fixed $a_0 > 0$, this expression is strictly increasing in b_a , and at equality its value is the rational number

$$\xi_{r, \ell} := \Omega_s(a_0, a_0) = \frac{a_0}{2^s} = \frac{\kappa_{q_{\text{und}}, s} \rho_{\text{und}}}{2^s}. \quad (8.6)$$

The preceding formulas give exactly the dichotomy in (8.5). No comparison between s and ℓ is required in this case. □

8.3 Undecidability and the comparison corollary

Proof of Theorem 1.2. Apply Proposition 8.1. For every fixed pair (r, ℓ) in the stated range, the map $a \mapsto \Psi(r, \ell, a)$ is total, its threshold $\Xi(r, \ell) = \xi_{r, \ell}$ is independent of a , and all outputs satisfy the promised lower bound.

Fix standard effective encodings $\langle \mathcal{F} \rangle$ of finite hypergraph families and $\langle \mathcal{H}, \mathcal{J} \rangle$ of ordered pairs of such families. For $\diamond \in \{>, =, \leq\}$, let

$$L_\diamond := \left\{ \langle \mathcal{F} \rangle : \pi_\ell^{(r)}(\mathcal{F}) \diamond \xi_{r, \ell} \right\}.$$

For the fixed pair (r, ℓ) , the total computable map $a \mapsto \langle \Psi(r, \ell, a) \rangle$ gives the many-one reductions

$$\mathfrak{U} \leq_m L_>, \quad \overline{\mathfrak{U}} \leq_m L_=: \quad \overline{\mathfrak{U}} \leq_m L_\leq.$$

Consequently, a decider for any one of the three languages would decide $\mathfrak{U}$, which is impossible. Every output of the reductions satisfies the promise $\pi_\ell^{(r)}(\mathcal{F}_a) \geq \xi_{r, \ell}$. This proves the versions with the promise. □

Proof of Corollary 1.3. It remains to construct one fixed finite family whose density is the threshold. If $\ell = 1$, apply the power construction in ordinary uniformity r with $h = 1$ to the constant score table g° , and let $\mathcal{G}_{r, \ell}$ be its pair-covering extension family. By Lemma 2.1 and Theorem 7.1,

$$\pi_1^{(r)}(\mathcal{G}_{r, \ell}) = \pi(\mathcal{G}_{r, \ell}) = \rho_{r, 1} = \xi_{r, 1}.$$

If $\ell \geq 2$ and $s < s_{\text{und}}$, apply the power construction to the constant score table g° with any fixed even denominator and then apply the clean-set transfer. The resulting finite family $\mathcal{G}_{r, \ell}$ has density (8.4).

If $\ell \geq 2$ and $s \geq s_{\text{und}}$, suspend two copies of the fixed ordinary equality class obtained by applying the construction to g° in the fixed uniformity q_{und} , again with a fixed even denominator. Applying the two-marker transfer to the resulting one-marker families gives a fixed family of density (8.6); denote it by $\mathcal{G}_{r, \ell}$.

Consequently the pairs $(\mathcal{F}_a, \mathcal{G}_{r, \ell})$ satisfy

$$\pi_\ell^{(r)}(\mathcal{F}_a) \geq \pi_\ell^{(r)}(\mathcal{G}_{r, \ell}),$$

with equality exactly when $a \notin \mathfrak{U}$ and strict inequality exactly when $a \in \mathfrak{U}$. Put $\delta(\mathcal{H}) := \pi_\ell^{(r)}(\mathcal{H})$. For the fixed pair (r, ℓ) , the total computable map $a \mapsto \langle \mathcal{F}_a, \mathcal{G}_{r, \ell} \rangle$ therefore gives the many-one reductions

$$\mathfrak{U} \leq_m \{ \langle \mathcal{H}, \mathcal{J} \rangle : \delta(\mathcal{H}) > \delta(\mathcal{J}) \},$$

$$\overline{\mathfrak{U}} \leq_m \{ \langle \mathcal{H}, \mathcal{J} \rangle : \delta(\mathcal{H}) = \delta(\mathcal{J}) \}, \quad \overline{\mathfrak{U}} \leq_m \{ \langle \mathcal{H}, \mathcal{J} \rangle : \delta(\mathcal{H}) \leq \delta(\mathcal{J}) \}.$$

Interchanging the two coordinates gives the corresponding reverse-order relations. $\square$

9 Effectivity and scope

Finite algorithms.

We use “effective for all sufficiently large parameters” in the sense fixed in the Introduction. This section lists the finite procedures, the results used from other papers, and the order in which the objects are chosen.

- (E1) For the undecidability reduction, fix the index of $\mathfrak{U}$, the universal polynomial P , and the stringent graph H used in [Theorem 6.3](#). On input a , the symbolic construction

$$P(a, Z) \mapsto p_a \mapsto q_a \mapsto \tau_\Delta(q_a) \mapsto \mathbf{L}(q_a) \mapsto (b_{a, F})_F \mapsto (g_a, N_a)$$

uses only finite polynomial and finite graph-algebra operations, and therefore computes the rational score table and an even common denominator. Independently, [Theorem 6.4](#) computes r_{tr} and, for every $r \geq r_{\text{tr}}$, the edge lists of $\mathcal{A}_r^{\text{tr}}$ and $\mathcal{P}_r^{\text{tr}}$ used by the transcendence construction.

- (E2) In the power construction, finite fields and the translation relations are constructed from N_a ; the spectral relation is found by finite search; the integer u , a Baranyai factorization, the balancing relations, and the correction supports are found by exact finite searches through explicitly bounded sets.
- (E3) The rooted admissibility predicate is decidable. The bounded witness tree in [Lemma 7.9](#) gives an explicit order bound; all nonmembers up to this bound and all pair-covering extensions are then enumerated.
- (E4) The clean-set and two-marker families are obtained by enumerating finitely many certificate assignments and all admissible quotients of their finite formal configurations.
- (E5) The suspension basis is obtained from the bounded common-root and link witness in [Proposition 4.1](#), followed by pair-covering extensions.
- (E6) [Lemma 7.17](#) gives the range in which the rational-power construction works, while [Lemma 4.5](#) computes $s_0(q)$ from the bound $(q^q/q!)(3/4)^s$. After choosing the two fixed pairs $(q_{\text{und}}, s_{\text{und}})$ and $(q_{\text{tr}}^*, s_{\text{tr}}^*)$, the finite constructions give the partial computable maps below. Equations (8.1)–(8.3) and [Lemma 2.1](#) prove that these maps are correct whenever $r \geq r_0$ and $1 \leq \ell \leq r - 1$. The maps Ξ , Ψ , and Φ_{cmp} use only the pair chosen for undecidability and do not call [Theorem 6.4](#). For $\ell = 1$, the maps use the ordinary construction directly; no clean-set or two-marker certificate is constructed.

Map (input)	Range of validity	Output
$\Phi_{\text{tr}}(r, \ell)$	halts for every pair in the stated range	a finite family with transcendental density
$\Xi(r, \ell)$	halts for every pair in the stated range	the exact rational threshold $\xi_{r, \ell}$
$\Psi(r, \ell, a)$	halts for every pair in the stated range, for all $a \in \mathbb{N}$	the reduction family $\mathcal{F}_a$
$\Phi_{\text{cmp}}(r, \ell)$	halts for every pair in the stated range	the comparison family $\mathcal{G}_{r, \ell}$

All four statements hold for every pair in the stated range. The last column gives proved properties of the outputs, not density tests performed by the maps. All family outputs are explicit finite incidence lists, and $\Xi(r, \ell)$ is computed by exact rational arithmetic. For fixed (r, ℓ) , the threshold $\Xi(r, \ell)$ is independent of a .

The probabilistic arguments prove existence of asymptotically extremal finite constructions; they are not needed by the algorithm that outputs a forbidden family.

Scope.

We do not optimize the uniformity threshold. The theorem concerns finite forbidden families, not single forbidden hypergraphs. It does not claim the same rational threshold works for different pairs (r, ℓ) . Within each fixed-pair decision problem, the threshold is fixed and neither r nor ℓ is included in the input encoding $\langle \mathcal{F} \rangle$; both remain inputs to the uniform construction maps above.

The proof of Lemma 7.17 gives a computable sufficient uniformity threshold through the finite comparisons in (7.52a)–(7.59). Together with Lemma 4.5 and Theorem 6.4, this gives a computable sufficient choice of r_0 . The construction algorithms do not require this threshold as input: they search directly for the required finite objects and halt throughout the stated range. We make no attempt to determine the least valid threshold or to optimize the numerical bounds.

Acknowledgements

H.L. was supported by the National Natural Science Foundation of China (12501487), the China Scholarship Council, and the Institute for Basic Science (IBS-R029-C4). X.L. was supported by the Excellent Young Talents Program (Overseas) of the National Natural Science Foundation of China.

Declaration on the use of AI

The authors used generative AI tools to discuss the organization of the proofs, check proof details, and improve the exposition.

References

- [1] J. Ai, L. Ding, H. Liu, and H. Yang. Tree suspensions and transfer functions for single degree Turán spectra. [arXiv:2607.06518](https://arxiv.org/abs/2607.06518), 2026.
- [2] R. Baber and J. Talbot. New Turán densities for 3-graphs. *Electron. J. Combin.*, 19(2):Paper 22, 21, 2012.
- [3] Z. Baranyai. On the factorization of the complete uniform hypergraph. In *Infinite and finite sets (Colloq., Keszthely, 1973; dedicated to P. Erdős on his 60th birthday)*, Vol. I, volume 10 of *Colloq. Math. Soc. János Bolyai*, pages 91–108. North-Holland, Amsterdam, 1975.
- [4] S. Basu, R. Pollack, and M.-F. Roy. *Algorithms in real algebraic geometry*, volume 10 of *Algorithms and Computation in Mathematics*. Springer-Verlag, Berlin, second edition, 2006.
- [5] B. Bollobás. Relations between sets of complete subgraphs. In *Proceedings of the Fifth British Combinatorial Conference (Univ. Aberdeen, Aberdeen, 1975)*, volume XV of *Congress. Numer.*, pages 79–84, Winnipeg, MB, 1976. Utilitas Math.
- [6] C. Bordenave. A new proof of Friedman’s second eigenvalue theorem and its extension to random lifts. *Ann. Sci. Éc. Norm. Supér. (4)*, 53(6):1393–1439, 2020.
- [7] G. Brito, I. Dumitriu, and K. D. Harris. Spectral gap in random bipartite biregular graphs and applications. *Combin. Probab. Comput.*, 31(2):229–267, 2022.
- [8] W. G. Brown and M. Simonovits. Digraph extremal problems, hypergraph extremal problems, and the densities of graph structures. *Discrete Math.*, 48(2-3):147–162, 1984.
- [9] M. Davis, H. Putnam, and J. Robinson. The decision problem for exponential Diophantine equations. *Ann. of Math. (2)*, 74:425–436, 1961.
- [10] P. Frankl and Z. Füredi. Extremal problems whose solutions are the blowups of the small Witt-designs. *J. Combin. Theory Ser. A*, 52(1):129–147, 1989.
- [11] P. Frankl and V. Rödl. Hypergraphs do not jump. *Combinatorica*, 4(2-3):149–159, 1984.

- [12] J. Friedman. A proof of Alon’s second eigenvalue conjecture and related problems. *Mem. Amer. Math. Soc.*, 195(910):viii+100, 2008.
- [13] Z. Füredi. Turán type problems. In *Surveys in combinatorics, 1991 (Guildford, 1991)*, volume 166 of *London Math. Soc. Lecture Note Ser.*, pages 253–300. Cambridge Univ. Press, Cambridge, 1991.
- [14] J. Gao, O. Pikhurko, M. Rong, and S. Sun. Rational codegree Turán density of hypergraphs. [arXiv:2601.00758](https://arxiv.org/abs/2601.00758), 2026.
- [15] C. Grosu. On the algebraic and topological structure of the set of Turán densities. *J. Combin. Theory Ser. B*, 118:137–185, 2016.
- [16] H. Hatami and S. Norine. Undecidability of linear inequalities in graph homomorphism densities. *J. Amer. Math. Soc.*, 24(2):547–565, 2011.
- [17] W. Hoeffding. Probability inequalities for sums of bounded random variables. *J. Amer. Statist. Assoc.*, 58(301):13–30, 1963.
- [18] J. P. Jones. Universal Diophantine equation. *J. Symbolic Logic*, 47(3):549–571, 1982.
- [19] G. Katona, T. Nemetz, and M. Simonovits. On a problem of Turán in the theory of graphs. *Mat. Lapok*, 15:228–238, 1964.
- [20] P. Keevash. Hypergraph Turán problems. In *Surveys in combinatorics 2011*, volume 392 of *London Math. Soc. Lecture Note Ser.*, pages 83–139. Cambridge Univ. Press, Cambridge, 2011.
- [21] H. Li, H. Liu, and X. Liu. Transcendence and undecidability in hypergraph Turán densities. Preprint, version dated 4 August 2026. Available at: https://github.com/xliu2022/xliu2022.github.io/tree/main/Unpolished_Manuscripts.
- [22] X. Liu and O. Pikhurko. Hypergraph Turán densities can have arbitrarily large algebraic degree. *J. Combin. Theory Ser. B*, 161:407–416, 2023.
- [23] X. Liu and O. Pikhurko. Intervals of hypergraph Turán densities. [arXiv:2605.25914](https://arxiv.org/abs/2605.25914), 2026.
- [24] A. Lo and K. Markström. ℓ -degree Turán density. *SIAM J. Discrete Math.*, 28(3):1214–1225, 2014.
- [25] L. Lovász. *Large networks and graph limits*, volume 60 of *American Mathematical Society Colloquium Publications*. American Mathematical Society, Providence, RI, 2012.
- [26] L. Lovász and B. Szegedy. Limits of dense graph sequences. *J. Combin. Theory Ser. B*, 96(6):933–957, 2006.
- [27] J. V. Matijasevič. The Diophantineness of enumerable sets. *Dokl. Akad. Nauk SSSR*, 191:279–282, 1970. Russian; English translation in *Soviet Math. Dokl.* 11 (1970), 354–358.
- [28] Y. V. Matiyasevich. *Hilbert’s tenth problem*. Foundations of Computing Series. MIT Press, Cambridge, MA, 1993.
- [29] T. S. Motzkin and E. G. Straus. Maxima for graphs and a new proof of a theorem of Turán. *Canadian J. Math.*, 17:533–540, 1965.
- [30] D. Mubayi, O. Pikhurko, and B. Sudakov. Hypergraph Turán problem: some open questions, 2011. AIM workshop problem list.
- [31] D. Mubayi and C. Terry. An extremal graph problem with a transcendental solution. *Combin. Probab. Comput.*, 28(2):303–324, 2019.
- [32] D. Mubayi and Y. Zhao. Co-degree density of hypergraphs. *J. Combin. Theory Ser. A*, 114(6):1118–1132, 2007.
- [33] O. Pikhurko. On possible Turán densities. *Israel J. Math.*, 201(1):415–454, 2014.
- [34] A. A. Razborov. Flag algebras. *J. Symbolic Logic*, 72(4):1239–1282, 2007.
- [35] A. Sidorenko. What we know and what we do not know about Turán numbers. *Graphs Combin.*, 11(2):179–199, 1995.
- [36] A. F. Sidorenko. Asymptotic solution for a new class of forbidden r -graphs. *Combinatorica*, 9(2):207–215, 1989.
- [37] J. Talbot. Lagrangians of hypergraphs. *Combin. Probab. Comput.*, 11(2):199–216, 2002.

- [38] B. Wu. An irrational Turán density via hypergraph Lagrangian densities. *Electron. J. Combin.*, 29(3):Paper No. 3.62, 11, 2022.
- [39] Z. Yan and Y. Peng. An irrational Lagrangian density of a single hypergraph. *SIAM J. Discrete Math.*, 36(1):786–822, 2022.
- [40] R. Yuster. On the maximum density of a matrix and a transcendental Turán-type density. [arXiv:2601.14904](https://arxiv.org/abs/2601.14904), 2026.